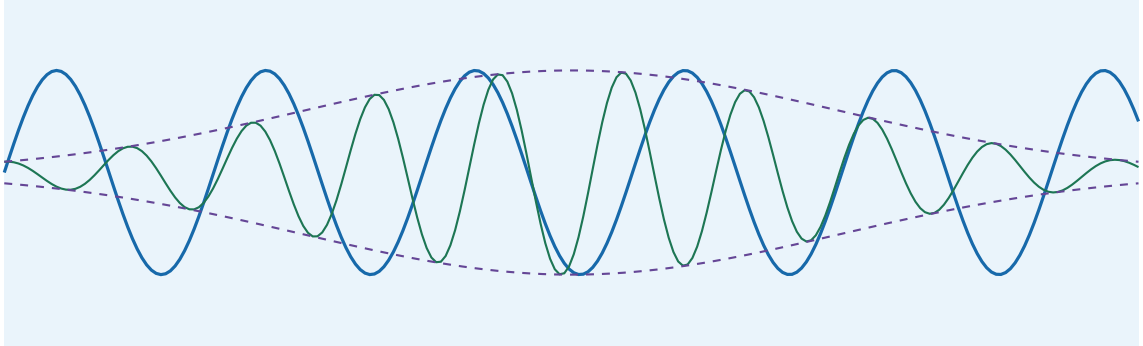
WAVES

Complete Learning Notes

Oscillators, mechanical waves, and light

Version V4.0 October 19, 2026

Organized by Shang Wu



Definition: Learning sequence

Restore equilibrium → diagonalize coupled motion → take the continuum limit → propagate energy → connect light to matter.

Full derivations, physical interpretation, annotated diagrams, and worked examples organized around all eight syllabus units.

Based on the supplied Waves syllabus and the relevant sections of Peatross and Ware, *Physics of Light and Optics*, May 15, 2024 revision. All instructional prose, worked examples, and figures in this edition are newly prepared.

Course Map and Reading Guide

Definition: What this edition covers

The required scope is the supplied two-page *Waves – Syllabus*: eight units from single oscillators through electromagnetic interference. Peatross and Ware provide the optical reference, not a replacement syllabus. Their ray optics, imaging, diffraction, holography, and blackbody chapters are not included as compulsory topics here.

Unit	Required coverage	Location and practice
1	Free, damped, and forced oscillators; phase; initial data; power; quality factor; phasors	Chapter 2; Problems O1–O8. Mathematical tools in Chapter 1.
2	Longitudinal and transverse spring coupling; slinky limit; two masses/three springs; modes and beats	Chapter 3; C1–C8.
3	Many degrees of freedom; dispersion; periodic boundaries; monatomic and binary lattices; forced response	Chapter 4; L1–L8.
4	String wave equation; transverse standing modes; boundary conditions; Fourier superposition	Chapter 5; S1–S8.
5	Traveling waves; phase and group speeds; energy flux; impedance; reflection and transmission	Chapter 6; T1–T8.
6	Maxwell to wave equation; plane waves; polarization; impedance; Poynting theorem; absorption	Chapter 7 and polarization in Chapter 9; E1–E8 and I2–I3.
7	Lorentz oscillator; complex susceptibility; absorption and phase speed in a medium	Chapter 8; M1–M8.
8	Interference; polarization overlap; visibility; unequal frequencies; group velocity; electromagnetic standing waves	Chapter 9; I1–I8.

How the supplied textbook is used

Page numbers below are the book's printed pages, not the PDF viewer counter (the main text has an eight-page offset).

- Chapter 0, Sections 0.1–0.4 (pp. 1–18): complex numbers, vector calculus, linear algebra, and Fourier tools.
- Sections 1.6–1.7 (pp. 34–38): polarization sources and the electromagnetic wave equation.
- Sections 2.1–2.4, 2.6–2.7 (pp. 43–51, 54–58): plane waves, complex refractive index, Lorentz model, energy transport. Section 2.5 (pp. 52–53) is used only for a labeled conductor/plasma extension.
- Chapter 6, Sections 6.1–6.6 (pp. 144–152): polarization notation and optional Jones-matrix tools.
- Chapter 7, Sections 7.1–7.5 (pp. 170–182): interference, beats, spectra, packets, and dispersion. A full coherence-theory chapter is not assumed to be part of the required syllabus.

The syllabus names Berkeley volumes I and III for Units 1–5 and Prof. Moti Segev's lecture notes for electromagnetism. These materials were not supplied. The mechanical chapters therefore provide self-contained derivations aligned to the syllabus; they are not presented as a page-by-page reconstruction

of unavailable lectures. No recent course examinations or tutorials were supplied, so no school-specific frequency ranking is asserted.

Standard Workflow: A route through the subject

For each model, identify the coordinates and approximations, derive the equation, solve its eigenvalue or response problem, impose the data or boundaries, and finish with an energy or limiting-case check. The same sequence reappears from two masses to a dielectric.

Color key

Blue: definitions and assumptions. Green: governing results and complete solutions. Purple: worked examples. Gold: standard workflows. Red: error checks. Gray: supplementary material beyond the explicit syllabus wording.

Contents

Course Map and Reading Guide i

1 Tools, Conventions, and the Physical Strategy 1

1.1 From a physical model to a solvable equation . . . 1
1.2 One phasor convention throughout . . . 1
1.3 Linear algebra and mass-weighted orthogonality . . . 1
1.4 Fourier and vector-calculus essentials . . . 2

2 Single Oscillators: Motion, Response, and Energy 2

2.1 Free motion and initial conditions . . . 3
2.2 Damping: three regimes and an exact initial-value solution . . . 3
2.3 Forced response: amplitude and phase are one complex number . . . 4
2.4 Which resonance is being asked for? . . . 5
2.5 Linearity, resonance without damping, and a standard answer . . . 5

3 Coupled Oscillators and Normal Coordinates 6

3.1 Two masses and three longitudinal springs . . . 6
3.2 Symmetric modes and modal initial data . . . 6
3.3 Transverse coupling, pre-tension, and the slinky limit . . . 7
3.4 Driving selects modes, not just frequencies . . . 8

4 Many Degrees of Freedom, Dispersion, and Phonons 8

4.1 Monatomic chain: derive the dispersion before interpreting it . . . 8
4.2 Boundary conditions count the allowed modes . . . 9
4.3 Continuum limit and sound . . . 9
4.4 Binary chain: define the cell before writing the phase . . . 10
4.5 Forced many-coordinate motion and mode selection . . . 11

5 String Waves, Boundaries, and Fourier Synthesis **11**
5.1 The string equation is a force balance . . . 11
5.2 Separation and endpoint conditions . . . 11
5.3 Arbitrary displacement and velocity data . . . 12
5.4 Mode energy and Parseval's physical meaning . . . 13
5.5 Driven strings and resonance selection . . . 14

6 Traveling Waves, Energy Flux, and Scattering **14**
6.1 d'Alembert's solution and propagation direction . . . 14
6.2 Energy density and directed power . . . 15
6.3 An interface: derive amplitude coefficients from continuity . . . 15
6.4 A localized barrier: a point mass . . . 16
6.5 Packets, phase speed, and group speed . . . 16

7 Electromagnetic Waves and Energy Transport **17**
7.1 Maxwell's equations and the material assumptions . . . 17
7.2 Derive the wave equation, including the source terms . . . 17
7.3 Poynting's theorem from Maxwell's equations . . . 18
7.4 Absorption and complex wave impedance . . . 19
7.5 A useful boundary connection . . . 19

8 The Lorentz Model: From a Driven Electron to Optical Response **20**
8.1 Microscopic displacement and macroscopic polarization . . . 20
8.2 From susceptibility to index, phase speed, and absorption . . . 21
8.3 Energy transfer explains the imaginary part . . . 21
8.4 Low, high, and resonant frequency limits . . . 22

9 Polarization, Interference, and Wave Packets **23**
9.1 Polarization is a trajectory, not a propagation direction . . . 23
9.2 Projection and optional matrix tools . . . 23
9.3 Derive the two-beam interference term . . . 24
9.4 Spatial fringes and optical path . . . 24
9.5 Unequal frequencies and the detector timescale . . . 25
9.6 Group index, delay, and pulse spreading . . . 25
9.7 Electromagnetic standing waves: fields and flux . . . 26

10 Integrated Revision and Formula Map **26**
10.1 One mathematical structure across the course . . . 26
10.2 Derivation checklist by syllabus unit . . . 26
10.3 A concise equation index . . . 27
10.4 An integrated worked application . . . 28
10.5 A complete, assessable answer . . . 28
10.6 Source and production record . . . 28

1 Tools, Conventions, and the Physical Strategy

Source map: Syllabus Units 1–8; Peatross and Ware Sections 0.1–0.4.

1.1 From a physical model to a solvable equation

An oscillation is repeated motion near equilibrium; a wave is organized motion distributed over space. Their common mathematical core is a linear restoring mechanism. A single displacement obeys an ordinary differential equation. A set of displacements gives a matrix equation. A continuum gives a partial differential equation. A sinusoidal trial solution converts each into an algebraic problem.

Definition: The linear regime

Expand a smooth potential near a stable equilibrium q_0:

$$U(q_0+\eta) = U(q_0) + U'(q_0)\eta + \tfrac{1}{2}U''(q_0)\eta^2 + O(\eta^3).$$

At equilibrium $U'(q_0) = 0$; stability requires $U''(q_0) > 0$ in the nondegenerate one-coordinate case. Neglecting higher orders gives $m\ddot{\eta} + U''(q_0)\eta = 0$. This is a local approximation, not a statement that every large-amplitude motion is harmonic.

1.2 One phasor convention throughout

Write a real harmonic quantity as

$$q(t) = \mathrm{Re}\{Q\mathrm{e}^{-\mathrm{i}\omega t}\}, \qquad Q = A\mathrm{e}^{\mathrm{i}\phi}, \qquad A \geq 0.$$

Then $q = A\cos(\omega t - \phi)$, $\dot{q} \leftrightarrow -\mathrm{i}\omega Q$, and $\ddot{q} \leftrightarrow -\omega^2 Q$. A positive argument of Q means a positive time lag relative to a cosine drive. For a forward spatial wave use $\mathrm{e}^{\mathrm{i}(kx-\omega t)}$; its complex amplitude includes $\mathrm{e}^{\mathrm{i}kx}$.

Key Result: Average a product, not the factors separately

If $a = \mathrm{Re}(A\mathrm{e}^{-\mathrm{i}\omega t})$ and $b = \mathrm{Re}(B\mathrm{e}^{-\mathrm{i}\omega t})$, then

$$\langle ab \rangle = \tfrac{1}{2}\,\mathrm{Re}(AB^*).$$

Proof: insert $a = (A\mathrm{e}^{-\mathrm{i}\omega t} + A^*\mathrm{e}^{\mathrm{i}\omega t})/2$, multiply, and discard the terms at 2ω. In particular $\langle a^2 \rangle = |A|^2/2$, whereas $\langle a \rangle = 0$.

Worked Example: Phase and power from a complex amplitude

Let $F(t) = 6\cos(4t)\,\mathrm{N}$ and $X = (0.03 + 0.04\mathrm{i})\,\mathrm{m}$. Then $|X| = 0.05\,\mathrm{m}$ and $\phi = \mathrm{atan2}(0.04, 0.03) = 53.13°$. Thus $x = 0.05\cos(4t - 53.13°)$. The velocity phasor is $V = -4\mathrm{i}X = 0.16 - 0.12\mathrm{i}\,\mathrm{m\,s^{-1}}$, so

$$\langle F\dot{x} \rangle = \tfrac{1}{2}\,\mathrm{Re}(6V^*) = 0.48\,\mathrm{W}.$$

Multiplying the real parts of the two phasors would not give the required average.

1.3 Linear algebra and mass-weighted orthogonality

For conservative coordinates $\boldsymbol{q}$, let M be a symmetric positive-definite mass matrix and K the symmetric stiffness matrix. A mode $\boldsymbol{q} = \boldsymbol{a}\cos\omega t$ satisfies

$$K\boldsymbol{a} = \omega^2 M\boldsymbol{a}.$$

Two eigenvectors with different ω^2 obey $\boldsymbol{a}_r^T M \boldsymbol{a}_s = 0$: transpose the two eigenvalue equations and subtract, using symmetry. The natural inner product is therefore $\langle \boldsymbol{u}, \boldsymbol{v} \rangle_M = \boldsymbol{u}^T M \boldsymbol{v}$, not always the

ordinary dot product. With $\boldsymbol{z} = M^{1/2}\boldsymbol{q}$, the matrix $D = M^{-1/2}KM^{-1/2}$ is symmetric and diagonalizable by an orthogonal matrix.

Standard Workflow: Modal initial-value problems

Find the eigenpairs, normalize $\boldsymbol{a}_r^T M\boldsymbol{a}_s = \delta_{rs}$, project $Q_r(0) = \boldsymbol{a}_r^T M\boldsymbol{q}(0)$ and $\dot{Q}_r(0) = \boldsymbol{a}_r^T M\dot{\boldsymbol{q}}(0)$, then reconstruct $\boldsymbol{q} = \sum_r \boldsymbol{a}_r Q_r(t)$. A degeneracy allows a choice of orthogonal basis within the degenerate subspace; it does not remove modes.

1.4 Fourier and vector-calculus essentials

Orthogonality replaces coupled equations by independent coefficients. On $0 < x < L$,

$$\int_0^L \sin\frac{n\pi x}{L}\sin\frac{m\pi x}{L}\,\mathrm{d}x = \frac{L}{2}\delta_{nm}.$$

Our spatial Fourier transform pair is

$$\widehat{f}(k) = \int_{-\infty}^{\infty} f(x)\mathrm{e}^{-\mathrm{i}kx}\mathrm{d}x, \qquad f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{f}(k)\mathrm{e}^{\mathrm{i}kx}\mathrm{d}k.$$

For a plane wave, $\nabla \to \mathrm{i}\boldsymbol{k}$, $\partial_t \to -\mathrm{i}\omega$. Two identities needed later are

$$\nabla\times(\nabla\times\boldsymbol{E}) = \nabla(\nabla\cdot\boldsymbol{E}) - \nabla^2\boldsymbol{E},$$
$$\nabla\cdot(\boldsymbol{E}\times\boldsymbol{H}) = \boldsymbol{H}\cdot(\nabla\times\boldsymbol{E}) - \boldsymbol{E}\cdot(\nabla\times\boldsymbol{H}).$$

Error Check: Units and amplitudes

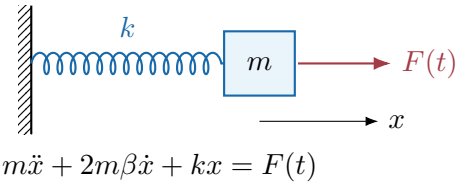
ω is in radians per second, while $f = \omega/(2\pi)$ is in hertz. Here $k = 2\pi/\lambda$ is angular wavenumber. RMS amplitude is peak amplitude divided by $\sqrt{2}$ for a scalar sinusoid. Using an RMS field in a formula already containing $1/2$ loses another factor of two.

Worked Example: Scaling without solving

If a string under tension $\mathcal{T}$ has linear density μ, the only speed made from them is $v = \sqrt{\mathcal{T}/\mu}$: $[\mathcal{T}] = \mathrm{kg\,m\,s^{-2}}$ and $[\mu] = \mathrm{kg\,m^{-1}}$. Dimensional analysis predicts this combination, but not the boundary-dependent factor $n\pi/L$ in a standing-mode frequency. Use dimensions as a check, not as a substitute for boundary conditions.

2 Single Oscillators: Motion, Response, and Energy

Source map: Syllabus Unit 1; self-contained mechanical derivations; phasor connection to Peatross and Ware Section 0.2.



$$m\ddot{x} + 2m\beta\dot{x} + kx = F(t)$$

Figure 2.1: A linear oscillator measured from equilibrium. The damping force is $-2m\beta\dot{x}$.

2.1 Free motion and initial conditions

Newton's law is $m\ddot{x} = -kx$, so $\omega_0 = \sqrt{k/m}$ and

$$x(t) = C\cos\omega_0 t + D\sin\omega_0 t = x_0\cos\omega_0 t + \frac{v_0}{\omega_0}\sin\omega_0 t.$$

Equivalently $x = A\cos(\omega_0 t - \phi)$, with $A = \sqrt{x_0^2 + (v_0/\omega_0)^2}$ and $\phi = \operatorname{atan2}(v_0/\omega_0, x_0)$. The phase is necessary to encode both initial position and velocity; a plain $A\cos\omega_0 t$ cannot fit arbitrary data.

Key Result: Energy of a free oscillator

$$E = \tfrac{1}{2}m\dot{x}^2 + \tfrac{1}{2}kx^2 = \tfrac{1}{2}kA^2, \qquad \langle T\rangle = \langle U\rangle = \frac{E}{2}.$$

Multiplying $m\ddot{x} + kx = 0$ by $\dot{x}$ gives $\dot{E} = 0$. The instantaneous kinetic and potential energies oscillate at $2\omega_0$ while their sum stays constant.

Worked Example: From a nonlinear force to a local oscillator

For $U(x) = ax^4 - bx^2$ with $a, b > 0$, equilibria satisfy $2x(2ax^2 - b) = 0$. At $x = 0$, $U'' = -2b < 0$, so small disturbances grow. At $x_* = \pm\sqrt{b/(2a)}$, $U'' = 12ax_*^2 - 2b = 4b$. Hence $\eta = x - x_*$ oscillates with $\omega_0 = 2\sqrt{b/m}$ for $|\eta| \ll |x_*|$. Stable equilibrium, rather than a globally quadratic potential, is the reason harmonic motion appears.

2.2 Damping: three regimes and an exact initial-value solution

With a velocity-proportional resistance,

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0, \qquad s^2 + 2\beta s + \omega_0^2 = 0.$$

Key Result: The three damping cases

$$\begin{aligned}
0 \le \beta < \omega_0: &\quad x = \mathrm{e}^{-\beta t}(C\cos\omega_d t + D\sin\omega_d t), \quad \omega_d = \sqrt{\omega_0^2 - \beta^2},\\
\beta = \omega_0: &\quad x = (C + Dt)\mathrm{e}^{-\omega_0 t},\\
\beta > \omega_0: &\quad x = C\mathrm{e}^{s_+ t} + D\mathrm{e}^{s_- t}, \quad s_\pm = -\beta \pm \sqrt{\beta^2 - \omega_0^2}.
\end{aligned}$$

For underdamping, $C = x_0$ and $D = (v_0 + \beta x_0)/\omega_d$. The extra βx_0 follows by differentiating the exponential envelope; omitting it violates the initial velocity.

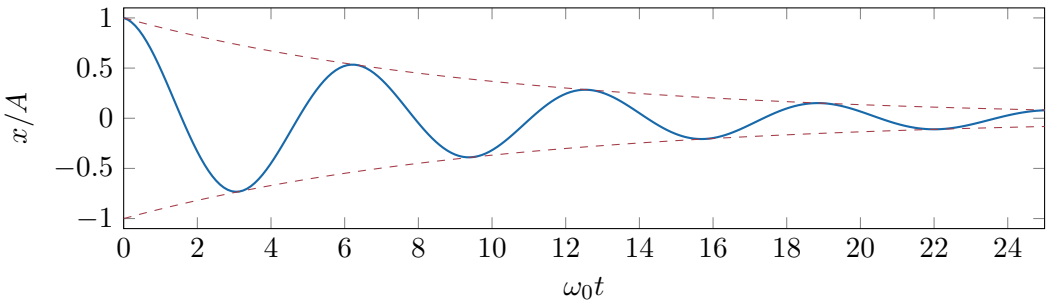


Figure 2.2: Underdamped motion and its displacement envelope. The energy envelope decays twice as fast as amplitude in weak damping.

The exact mechanical energy obeys

$$\dot{E} = m\dot{x}(\ddot{x} + \omega_0^2 x) = -2m\beta\dot{x}^2 \le 0.$$

Writing $x = Ae^{-\beta t}\cos\theta$, $\theta = \omega_d t - \phi$, gives the exact expression

$$E = \tfrac{1}{2}mA^2e^{-2\beta t}\{(\beta\cos\theta + \omega_d\sin\theta)^2 + \omega_0^2\cos^2\theta\}.$$

The braces are not constant. For $\beta \ll \omega_0$, a local cycle average changes little over one cycle and gives $\overline{E} \simeq \tfrac{1}{2}m\omega_0^2A^2e^{-2\beta t}$. Thus $\tau_A = 1/\beta$, $\tau_E = 1/(2\beta)$, and the weak-damping loss factor is $Q \simeq \omega_0/(2\beta)$. We use $Q = \omega_0/(2\beta)$ as the standard oscillator parameter even outside the regime where a sharp resonance exists.

Worked Example: Infer damping from ring-down data

An amplitude falls to 1/e of its initial envelope in 5.0 s; the measured oscillation frequency is $\omega_d = 10.0\,\mathrm{s}^{-1}$. Then $\beta = 0.200\,\mathrm{s}^{-1}$, $\omega_0 = \sqrt{100 + .0400} = 10.002\,\mathrm{s}^{-1}$, and $Q = 25.005$. The energy envelope reaches 1/e in 2.5 s. If successive same-sign peaks have ratio R, their logarithmic decrement is $\delta = \ln R = 2\pi\beta/\omega_d$ and $Q = \tfrac{1}{2}\sqrt{1 + (2\pi/\delta)^2} \simeq \pi/\delta$.

2.3 Forced response: amplitude and phase are one complex number

For a drive $F_0\cos\omega t$, take $x_{\mathrm{ss}} = \mathrm{Re}(Xe^{-\mathrm{i}\omega t})$. Substitution gives

$$m(\omega_0^2 - \omega^2 - 2\mathrm{i}\beta\omega)X = F_0.$$

Key Result: Transfer function and phase lag

$$X = \frac{F_0/m}{\omega_0^2 - \omega^2 - 2\mathrm{i}\beta\omega}, \qquad A = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}},$$

$$\phi = \mathrm{atan2}(2\beta\omega, \omega_0^2 - \omega^2), \qquad x_{\mathrm{ss}} = A\cos(\omega t - \phi).$$

The full solution is $x = x_{\mathrm{ss}} + x_{\mathrm{transient}}$; only the second term can impose the remaining initial data. A damped transient decays, leaving a response at the drive frequency, not at ω_d.

Regime	Amplitude	Lag
$\omega \ll \omega_0$	$A \simeq F_0/k$	$\phi \simeq 0$
$\omega = \omega_0$	$A = F_0/(2m\beta\omega_0)$	$\phi = \pi/2$
$\omega \gg \omega_0, \beta$	$A \simeq F_0/(m\omega^2)$	$\phi \to \pi$

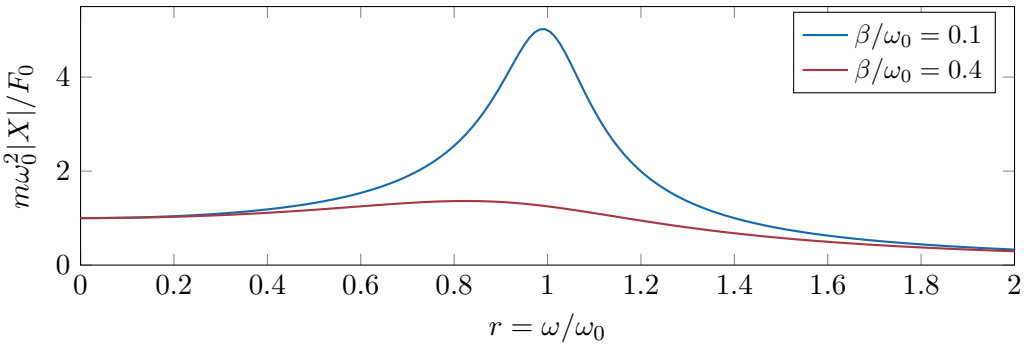


Figure 2.3: Displacement response for two damping strengths. Increased damping broadens and lowers the response.

2.4 Which resonance is being asked for?

Minimize $D(\omega) = (\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2$. Since $D' = 4\omega(\omega^2 - \omega_0^2 + 2\beta^2)$, a nonzero displacement maximum exists only if $\beta < \omega_0/\sqrt{2}$, at

$$\omega_{\text{amp}} = \sqrt{\omega_0^2 - 2\beta^2}.$$

For stronger damping, the displacement amplitude decreases from its static value; there is no nonzero-frequency displacement peak. This does not eliminate a peak in absorbed power.

Key Result: Mean absorbed power and energy balance

$$\overline{P} = \langle F\dot{x} \rangle = \frac{1}{2}F_0 A\omega \sin\phi = m\beta\omega^2 A^2 = \frac{\beta F_0^2\omega^2/m}{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}.$$

For frequency-independent viscous damping, $\overline{P}$ is maximal at $\omega = \omega_0$, with $P_{\max} = F_0^2/(4m\beta)$. The velocity amplitude peaks there too. The steady average energy satisfies input power = dissipated power.

At half power, divide the denominator by ω^2 to obtain $(\omega_0^2/\omega - \omega)^2 = 4\beta^2$. Thus

$$\omega_- = \sqrt{\omega_0^2 + \beta^2} - \beta, \quad \omega_+ = \sqrt{\omega_0^2 + \beta^2} + \beta, \quad \Delta\omega = 2\beta, \quad Q = \frac{\omega_0}{\Delta\omega}.$$

These are exact for the power curve of this model. The approximate displacement linewidth is the same only in the narrow-resonance limit.

Worked Example: Compare three frequencies in one experiment

Take $m = 1\,\text{kg}$, $\omega_0 = 5\,\text{s}^{-1}$, $\beta = 1\,\text{s}^{-1}$, $F_0 = 2\,\text{N}$. Free ring-down occurs at $\sqrt{24} = 4.899\,\text{s}^{-1}$; the displacement resonance is at $\sqrt{23} = 4.796\,\text{s}^{-1}$; maximum power is at $5\,\text{s}^{-1}$. At the last frequency, $A = 0.2\,\text{m}$, $\phi = 90°$, and $\overline{P} = 1\,\text{W}$. The half-power frequencies are $\sqrt{26} \mp 1$, whose difference is exactly $2\,\text{s}^{-1}$.

2.5 Linearity, resonance without damping, and a standard answer

If $F = \sum_j F_j \cos(\omega_j t + \alpha_j)$, add the individual responses. Do not combine different frequencies into one phasor. At zero damping and exact resonance, a bounded steady-state ansatz fails: for a drive turned on at rest,

$$x(t) = \frac{F_0}{2m\omega_0} t \sin\omega_0 t.$$

Substitution verifies the equation and initial data; the growing envelope explains why dividing by $\omega_0^2 - \omega^2$ at resonance is invalid.

Standard Workflow: Single-oscillator questions

Locate the equilibrium and linearize; divide by m to identify ω_0 and β; separate free/transient motion from driven steady motion; determine both amplitude and phase; apply initial data to the full solution; distinguish the requested resonance; check units and power loss. If damping is zero, treat exact resonance separately before dividing.

Error Check: A free oscillator does not have a frequency-response peak by itself

Its natural frequency is a property of the homogeneous motion. Resonance is a statement about response to a drive or energy transfer. In weak damping ω_d, ω_{amp}, and ω_0 are close, but they are not interchangeable in an exact derivation.

3 Coupled Oscillators and Normal Coordinates

Source map: Syllabus Unit 2; connections to generalized eigenvalue problems in Algebra 2M.

3.1 Two masses and three longitudinal springs

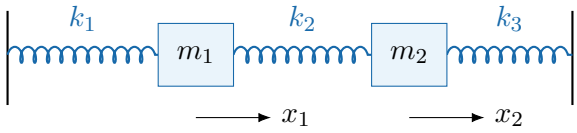


Figure 3.1: Displacements are measured from the static equilibrium configuration. The middle spring extension is $x_2 - x_1$.

The kinetic and quadratic potential energies are

$$T = \tfrac{1}{2}m_1\dot{x}_1^2 + \tfrac{1}{2}m_2\dot{x}_2^2, \qquad U = \tfrac{1}{2}k_1x_1^2 + \tfrac{1}{2}k_2(x_2 - x_1)^2 + \tfrac{1}{2}k_3x_2^2.$$

Taking $-\partial U/\partial x_j$ yields

$$M\ddot{\boldsymbol{x}} + K\boldsymbol{x} = 0, \quad M = \begin{pmatrix} m_1 & 0 \\ 0 & m_2 \end{pmatrix}, \quad K = \begin{pmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 + k_3 \end{pmatrix}.$$

The off-diagonal entries are negative because moving one mass right pulls its neighbor right rather than restoring that neighbor to the origin. A trial $\boldsymbol{x} = \boldsymbol{a}\mathrm{e}^{-\mathrm{i}\omega t}$ gives $\det(K - \omega^2 M) = 0$.

Key Result: The general two-frequency formula

With $A = (k_1 + k_2)/m_1$ and $B = (k_2 + k_3)/m_2$,

$$\omega_{\pm}^2 = \frac{A + B \pm \sqrt{(A - B)^2 + 4k_2^2/(m_1m_2)}}{2}.$$

The amplitude ratio follows from either row:

$$\frac{a_2}{a_1} = \frac{k_1 + k_2 - m_1\omega^2}{k_2} \quad (k_2 \neq 0).$$

If $k_2 = 0$, solve two independent oscillators; do not divide by zero. A zero eigenvalue describes an unrestrained coordinate, not an oscillation with finite period.

3.2 Symmetric modes and modal initial data

For $m_1 = m_2 = m$, $k_1 = k_3 = k$, and $k_2 = \kappa$,

$$\boldsymbol{a}_s = (1, 1)^T, \quad \omega_s^2 = k/m; \qquad \boldsymbol{a}_a = (1, -1)^T, \quad \omega_a^2 = (k + 2\kappa)/m.$$

In the symmetric mode, the middle spring is unextended. In the antisymmetric mode it changes length by twice the displacement, raising the frequency. Define $q_s = (x_1 + x_2)/2$ and $q_a = (x_1 - x_2)/2$. Adding and subtracting Newton's equations gives independent equations for q_s and q_a. These coordinates need not be mass-normalized to decouple; normalization matters when computing modal forces and energy.

Worked Example: A motion that is not one normal mode

Release $(x_1, x_2) = (A, 0)$ from rest. Then $q_s(0) = q_a(0) = A/2$ and

$$x_1 = \tfrac{A}{2}(\cos\omega_s t + \cos\omega_a t) = A\cos(\bar{\omega}t)\cos(\Delta\omega t/2),$$
$$x_2 = \tfrac{A}{2}(\cos\omega_s t - \cos\omega_a t) = A\sin(\bar{\omega}t)\sin(\Delta\omega t/2),$$

where $\bar{\omega} = (\omega_a + \omega_s)/2$ and $\Delta\omega = \omega_a - \omega_s > 0$. Both coordinates obey the original initial data. The slow exchange timescale is $\pi/\Delta\omega$ for the first maximum of the second displacement envelope. The intensity/energy-envelope repetition time is $2\pi/\Delta\omega$.

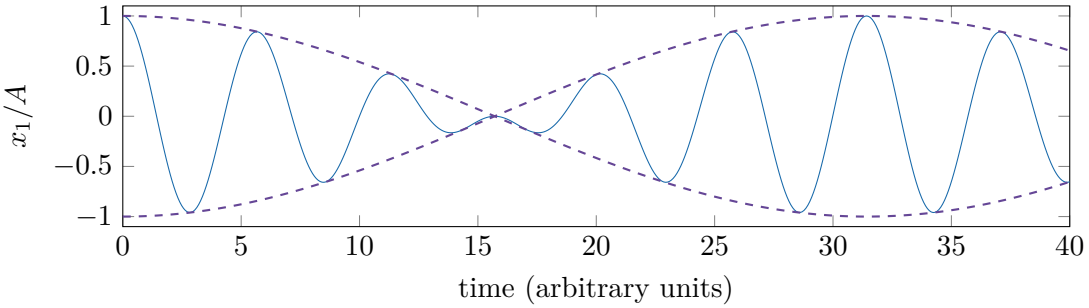


Figure 3.2: Two nearby mode frequencies produce a fast carrier and a slow envelope.

For weak coupling, $\omega_a \simeq \omega_s(1 + \kappa/k)$ and $\Delta\omega \simeq \kappa/(m\omega_s)$. A clean cycle-averaged transfer of local oscillator energy is then approximately described by $\cos^2(\Delta\omega t/2)$ and $\sin^2(\Delta\omega t/2)$. The coupling spring also stores energy. Exact total energy is conserved, but assigning all coupling energy to one mass and claiming exact independent energy fractions is generally wrong.

Worked Example: Unequal masses require a mass inner product

Let $M = \mathrm{diag}(m, 2m)$ and $K = k\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. With $s = m\omega^2/k$, the determinant is $(2-s)(2-2s) - 1 = 2s^2 - 6s + 3$. Thus $s_\pm = (3 \pm \sqrt{3})/2$. Choose $\boldsymbol{a}_\pm = (1, 2 - s_\pm)^T$. Their ordinary dot product is $1/2$, not zero, but

$$\boldsymbol{a}_+^T M \boldsymbol{a}_- = m\{1 + 2(2 - s_+)(2 - s_-)\} = 0.$$

To project initial data, divide each vector by $\sqrt{m[1 + 2(2 - s_\pm)^2]}$ first. This is the same self-adjoint structure used in linear algebra with a weighted inner product.

3.3 Transverse coupling, pre-tension, and the slinky limit

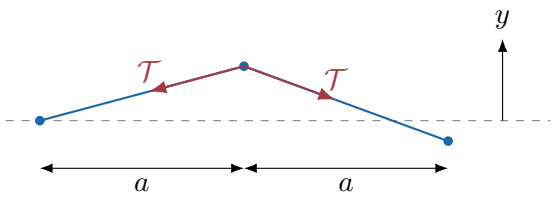


Figure 3.3: The vertical force is the difference of the vertical components of the tensions on neighboring segments.

Let equilibrium spacing be a and vertical displacements y_j. If the segment slopes satisfy $|y_{j+1} - y_j|/a \ll 1$ and the equilibrium tension is $\mathcal{T}$, then $\sin\theta \simeq \tan\theta = (y_{j+1} - y_j)/a$. Newton's law gives

$$m\ddot{y}_j = \frac{\mathcal{T}}{a}(y_{j+1} - 2y_j + y_{j-1}).$$

The effective transverse coupling is $\mathcal{T}/a$, not automatically the longitudinal spring stiffness. For a spring of stiffness K and natural length ℓ_0 stretched to a, $\mathcal{T} = K(a - \ell_0)$, so $K_\perp = K(1 - \ell_0/a)$. The change of tension due to a small transverse slope is second order and can be neglected in the linear force.

Key Result: Why the slinky limit needs a condition

If the equilibrium spring length is much greater than its natural length ($\ell_0/a \ll 1$), then $\mathcal{T} \simeq Ka$ and $K_\perp \simeq K$. Longitudinal and transverse equations then have the same coupling scale. Without pre-tension ($a = \ell_0$), the linear transverse restoring coefficient vanishes; the leading force is nonlinear. A taut-string equation is not justified for an unstretched chain simply by making the displacement small.

Worked Example: Two transverse masses

Three equal pre-tensioned segments of length a connect two equal masses to fixed endpoints. The equations are the equal-three-spring equations with $k = \kappa = \mathcal{T}/a$. Thus $\omega_s = \sqrt{\mathcal{T}/(ma)}$ and $\omega_a = \sqrt{3\mathcal{T}/(ma)}$. The mode shapes are $(1, 1)$ and $(1, -1)$, but the physical displacements are vertical. If the tension doubles at fixed a, both frequencies scale by $\sqrt{2}$.

3.4 Driving selects modes, not just frequencies

For mass-normalized modes, a force $\boldsymbol{f}(t)$ produces

$$\ddot{Q}_r + \omega_r^2 Q_r = \boldsymbol{a}_r^T \boldsymbol{f}(t).$$

A symmetric force (F, F) has zero projection on an antisymmetric mode. A resonance denominator alone does not guarantee a response: the corresponding forcing numerator can vanish. If damping is proportional to M, each mode acquires the same $2\beta\dot{Q}_r$; a general damping matrix need not diagonalize in the conservative mode basis.

Standard Workflow: Coupled-system questions

Draw the equilibrium geometry; express every spring extension; construct M and K; solve the generalized eigenvalue equation; calculate and normalize mode shapes; project the data or force; reconstruct the coordinates; check uncoupled, symmetric, and zero-frequency limits.

4 Many Degrees of Freedom, Dispersion, and Phonons

Source map: Syllabus Unit 3; self-contained discrete-chain derivations.

4.1 Monatomic chain: derive the dispersion before interpreting it

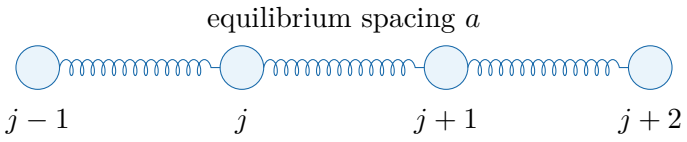


Figure 4.1: A nearest-neighbor longitudinal chain with mass m, spring stiffness K, and spacing a.

For displacement u_j from equilibrium,

$$m\ddot{u}_j = K(u_{j+1} - 2u_j + u_{j-1}).$$

Set $u_j = U\mathrm{e}^{\mathrm{i}(kja-\omega t)}$. The neighboring factors are $\mathrm{e}^{\pm \mathrm{i}ka}$, giving

$$-m\omega^2 = K(2\cos ka - 2) = -4K\sin^2(ka/2).$$

Key Result: Monatomic lattice dispersion

$$\omega(k) = 2\sqrt{K/m}\,|\sin(ka/2)|, \qquad -\pi/a \le k \le \pi/a.$$

Wavevectors differing by $2\pi/a$ give the same displacement at every site. The first Brillouin zone removes this redundancy; it is not a restriction that the wavelength of an arbitrary continuous function must satisfy.

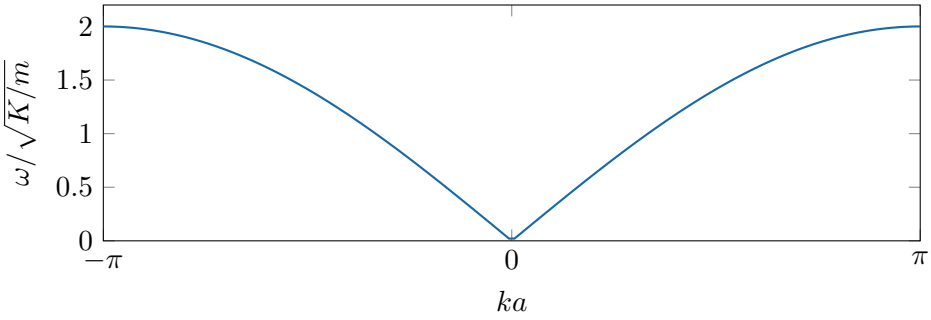


Figure 4.2: The lattice has a finite frequency band and a nonlinear dispersion relation away from long wavelengths.

For $0 < ka < \pi$,

$$v_p = \frac{\omega}{k}, \qquad v_g = \frac{\mathrm{d}\omega}{\mathrm{d}k} = a\sqrt{K/m}\cos(ka/2).$$

At $ka \ll 1$, $\omega \simeq c_s|k|$ with $c_s = a\sqrt{K/m}$. At the zone edge, $v_g = 0$ although $\omega \neq 0$: the mode oscillates, but a narrow packet centered there has zero first-order envelope velocity.

4.2 Boundary conditions count the allowed modes

Definition: Periodic and fixed-end chains

For N masses in a ring, $u_{j+N} = u_j$ implies $\mathrm{e}^{\mathrm{i}kNa} = 1$ and $k_s = 2\pi s/(Na)$, with N inequivalent values modulo $2\pi/a$. The $k = 0$ mode is uniform translation and has zero restoring force.
For N moving masses between fixed sites $u_0 = u_{N+1} = 0$, use

$$u_j^{(n)} \propto \sin\frac{n\pi j}{N+1}, \quad k_n = \frac{n\pi}{(N+1)a}, \quad n = 1, \dots, N.$$

There is no translational zero mode because the supports break translation symmetry.

For the fixed chain, discrete sine orthogonality gives $\sum_{j=1}^{N} \sin(k_n ja)\sin(k_m ja) = (N+1)\delta_{nm}/2$. This proves completeness of the N orthogonal eigenvectors, rather than merely guessing some modes.

Worked Example: A five-mass chain

Take $N = 5$, $K/m = 100\,\mathrm{s}^{-2}$. Then $\omega_n = 20\sin(n\pi/12)\,\mathrm{s}^{-1}$, giving approximately 5.176, 10.000, 14.142, 17.321, and 19.319. Mode $n = 3$ has site pattern $(1, 0, -1, 0, 1)$ up to a scale. The number of independent modes equals the number of moving coordinates, five, not the number of springs, six.

4.3 Continuum limit and sound

Replace $u_j(t)$ by $u(x,t)$ with $x = ja$. Taylor-expand:

$$u(x+a) - 2u(x) + u(x-a) = a^2 u_{xx} + \frac{a^4}{12}u_{xxxx} + \cdots.$$

Keeping only the first term gives $u_{tt} = c_s^2 u_{xx}$. This requires spatial variation over lengths much greater than a. The higher derivatives produce lattice dispersion and cannot be ignored for wavelengths of a few lattice spacings. For a transverse chain, substitute $K = \mathcal{T}/a$ and $m = \mu a$ to get $c_s^2 = \mathcal{T}/\mu$.

Worked Example: A dispersive packet

With $a = 0.010\,\mathrm{m}$, $K/m = 10^4\,\mathrm{s}^{-2}$, and carrier $ka = 2\pi/3$, $\omega = 100\sqrt{3}\,\mathrm{s}^{-1}$. Thus $v_p = 3\sqrt{3}/(2\pi) = 0.827\,\mathrm{m\,s}^{-1}$, whereas $v_g = 0.500\,\mathrm{m\,s}^{-1}$. The long-wave speed is $1.00\,\mathrm{m\,s}^{-1}$. A wave crest and a packet envelope do not generally move together.

4.4 Binary chain: define the cell before writing the phase

Let adjacent-site spacing be a and unit-cell length be $2a$. In cell ℓ, mass M at $2\ell a$ has displacement u_ℓ, and mass m at $(2\ell+1)a$ has displacement v_ℓ. Neighboring springs all have stiffness K:

$$M\ddot{u}_\ell = K(v_\ell + v_{\ell-1} - 2u_\ell),$$
$$m\ddot{v}_\ell = K(u_\ell + u_{\ell+1} - 2v_\ell).$$

Take $u_\ell = U\mathrm{e}^{\mathrm{i}(2\ell aq-\omega t)}$ and $v_\ell = V\mathrm{e}^{\mathrm{i}(2\ell aq-\omega t)}$. Both amplitudes use the cell origin; the position offset of the second atom is absorbed in V. The matrix is

$$\begin{pmatrix} 2K - M\omega^2 & -K(1+\mathrm{e}^{-2\mathrm{i}qa}) \\ -K(1+\mathrm{e}^{2\mathrm{i}qa}) & 2K - m\omega^2 \end{pmatrix} \begin{pmatrix} U \\ V \end{pmatrix} = 0.$$

Its determinant gives $Mm\omega^4 - 2K(M+m)\omega^2 + 4K^2\sin^2(qa) = 0$.

Key Result: Acoustic and optical branches

$$\omega_\pm^2 = K\left(\frac{1}{M}+\frac{1}{m}\right) \pm K\sqrt{\left(\frac{1}{M}+\frac{1}{m}\right)^2 - \frac{4}{Mm}\sin^2(qa)}, \quad |q| \le \frac{\pi}{2a}.$$

The minus sign is acoustic, with $\omega_-(0) = 0$ and $U = V$ at $q = 0$. The plus sign is optical, with $\omega_+^2(0) = 2K(1/M + 1/m)$ and $MU + mV = 0$ there. Optical means finite frequency at long wavelength, not necessarily visible light or guaranteed optical activity.

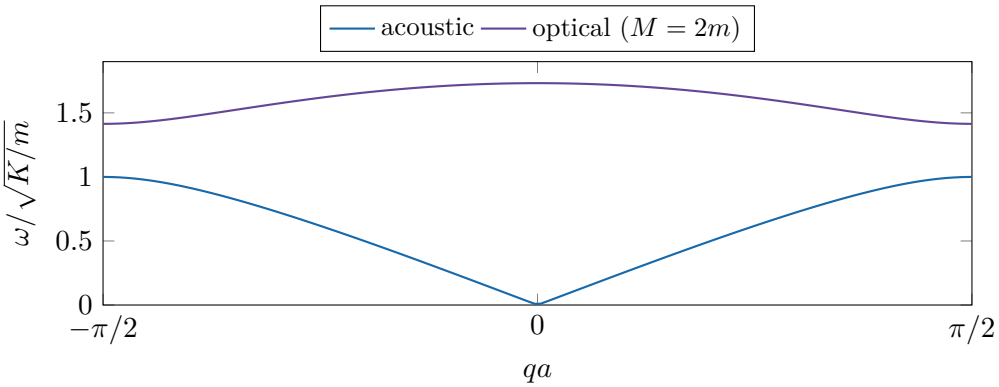


Figure 4.3: Two masses per cell produce two frequency branches. The gap is nonzero when the masses differ.

If $M > m$, the zone-edge limits are $\omega_-^2 = 2K/M$ and $\omega_+^2 = 2K/m$. No extended normal mode has frequency in the interval between these band edges. When $M = m$, the gap closes: this is the monatomic dispersion folded into a smaller Brillouin zone, not a new physical gap. At small q, expand the square root to find

$$\omega_- \simeq a\sqrt{\frac{2K}{M+m}}\,|q|.$$

Worked Example: Check the binary-chain limits

For $M = 3m$, the acoustic zone-edge frequency is $\sqrt{2K/(3m)}$, the optical edge is $\sqrt{2K/m}$, and the optical zone-center frequency is $\sqrt{8K/(3m)}$. At the optical center, $V = -3U$: the lighter mass moves more, keeping the cell's center of mass fixed. There are $2N_c$ modes for N_c cells, N_c on each branch.

4.5 Forced many-coordinate motion and mode selection

If $M\ddot{\boldsymbol{u}} + 2\beta M\dot{\boldsymbol{u}} + K\boldsymbol{u} = \mathrm{Re}(\boldsymbol{f}\mathrm{e}^{-\mathrm{i}\omega t})$, mass-normalized modes yield

$$\widetilde{Q}_r = \frac{\boldsymbol{a}_r^T \boldsymbol{f}}{\omega_r^2 - \omega^2 - 2\mathrm{i}\beta\omega}, \qquad \widetilde{\boldsymbol{u}} = \sum_r \boldsymbol{a}_r \widetilde{Q}_r.$$

A local drive excites every mode with nonzero displacement at the drive site. A spatially periodic drive selects matching wavevectors in a translationally invariant lattice. At an undamped modal resonance, use a growing resonant solution if the force projection is nonzero.

Supplementary Background: What is a phonon?

A normal mode is a classical collective coordinate. Upon quantization its energy levels are $E_r = \hbar\omega_r(n_r + 1/2)$, and one added quantum is a phonon. The lattice dispersion specifies phonon frequencies and group velocities. Quantization is a connection mentioned by the syllabus, not a change to the classical equations derived above.

Standard Workflow: Lattice problems

Specify site spacing and cell length; derive nearest-neighbor equations; insert a phase-consistent trial wave; solve the determinant; count modes using boundaries; distinguish branch from wavevector; differentiate the correct branch for v_g; check $q \to 0$, zone edge, and equal-mass limits.

5 String Waves, Boundaries, and Fourier Synthesis

Source map: Syllabus Unit 4; Fourier conventions linked to Peatross and Ware Section 0.4.

5.1 The string equation is a force balance

Let $y(x,t)$ be the small transverse displacement of a taut string, with tension $\mathcal{T}$ and mass per length μ. For an element $\mathrm{d}x$, the difference of vertical tension components is $\mathcal{T}[y_x(x+\mathrm{d}x) - y_x(x)]$. Its mass is $\mu\mathrm{d}x$, hence

$$\mu y_{tt} = \mathcal{T} y_{xx}, \qquad v = \sqrt{\mathcal{T}/\mu}.$$

We assumed $|y_x| \ll 1$, constant tension, negligible bending stiffness, and negligible damping. The disturbance is transverse, although the net energy travels along the string. A string particle does not move along the entire string with the wave speed.

5.2 Separation and endpoint conditions

Set $y = X(x)Q(t)$. Dividing $X\ddot{Q} = v^2 X'' Q$ by $v^2 XQ$ yields a constant $-k^2$:

$$X'' + k^2 X = 0, \qquad \ddot{Q} + \omega^2 Q = 0, \qquad \omega = vk.$$

The boundary conditions determine k; the differential equation alone does not quantize frequencies.

Key Result: Three common endpoint combinations

Ends	Spatial function	Allowed k
fixed–fixed	$\sin(kx)$	$n\pi/L$, $n = 1, 2, \ldots$
fixed–free	$\sin(kx)$	$(n + 1/2)\pi/L$, $n = 0, 1, \ldots$
free–free	$\cos(kx)$	$n\pi/L$, $n = 0, 1, \ldots$

A fixed end has $y = 0$; an ideal free end has $y_x = 0$, since the transverse force is zero. The free–free $n = 0$ mode is rigid transverse translation: its time dependence is $A + Bt$, not a nonzero-frequency sine.

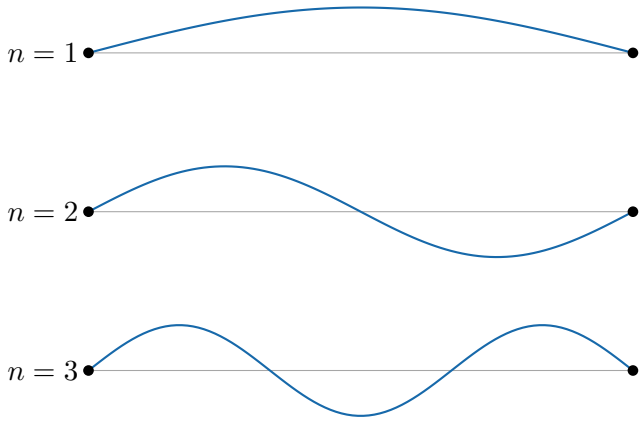


Figure 5.1: The first three fixed–fixed string shapes, each with zero endpoint displacement.

For fixed–fixed boundaries, $\lambda_n = 2L/n$ and $f_n = nv/(2L)$. There are $n - 1$ internal nodes. At a node the displacement is always zero; at an antinode the amplitude is maximal. A point that happens to cross $y = 0$ at one instant is not necessarily a node.

Worked Example: Infer tension from a mode pattern

A 1.20 m string of density $0.0050\,\mathrm{kg\,m^{-1}}$ has four loops at 200 Hz. Four loops means $n = 4$, so $\lambda = 2L/4 = 0.600\,\mathrm{m}$, $v = f\lambda = 120\,\mathrm{m\,s^{-1}}$, and $\mathcal{T} = \mu v^2 = 72.0\,\mathrm{N}$. Its fundamental is 50.0 Hz. Counting internal nodes as the mode number would give the wrong tension.

5.3 Arbitrary displacement and velocity data

For initial shape $f(x)$ and velocity $g(x)$ on a fixed string,

$$y(x,t) = \sum_{n=1}^{\infty} [A_n \cos\omega_n t + B_n \sin\omega_n t] \sin\frac{n\pi x}{L}.$$

Multiply the initial conditions by the mth sine and integrate. Orthogonality isolates one coefficient:

Key Result: Fourier coefficients for initial data

$$A_n = \frac{2}{L}\int_0^L f(x)\sin\frac{n\pi x}{L}\mathrm{d}x, \qquad B_n = \frac{2}{L\omega_n}\int_0^L g(x)\sin\frac{n\pi x}{L}\mathrm{d}x.$$

Displacement and velocity require separate projections. For mixed boundaries, use the corresponding half-integer sine basis; do not keep the fixed–fixed frequencies.

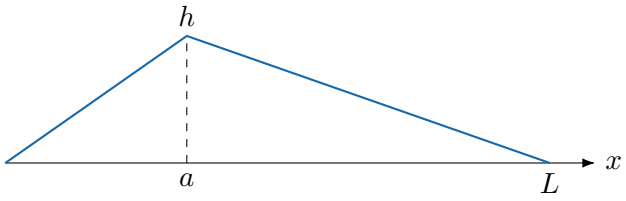


Figure 5.2: A triangular pluck at $x = a$. Its lack of smoothness populates many normal modes.

Worked Example: Derive the complete triangular-pluck spectrum

Let $f(x) = hx/a$ for $0 < x < a$ and $f(x) = h(L-x)/(L-a)$ for $a < x < L$, with zero initial velocity. Splitting the integral at a gives

$$A_n = \frac{2h}{L}\left[\frac{1}{a}\int_0^a x\sin k_n x\mathrm{d}x + \frac{1}{L-a}\int_a^L (L-x)\sin k_n x\mathrm{d}x\right]$$
$$= \frac{2hL^2}{\pi^2 n^2 a(L-a)}\sin\frac{n\pi a}{L}, \qquad B_n = 0.$$

The antiderivative $\int x\sin kx\mathrm{d}x = -x\cos kx/k + \sin kx/k^2$ explains the n^{-2} decay; the cosine endpoint terms cancel between segments. For a midpoint pluck, $A_n = 8h\sin(n\pi/2)/(\pi^2 n^2)$, so all even modes vanish. For a pluck at $L/3$, every third mode vanishes. The selection rule comes from the sine factor, not from a new dispersion relation.

Worked Example: Initial velocity excites a different quadrature

Let $y(x,0) = 0$ and $y_t(x,0) = V\sin(3\pi x/L)$. Then $A_n = 0$ and $B_n = (V/\omega_3)\delta_{n3}$. Therefore

$$y(x,t) = \frac{V}{\omega_3}\sin\omega_3 t\sin\frac{3\pi x}{L}.$$

At $t = 0$ the shape is flat even though the string is not at rest. Inferring zero energy from an instantaneous flat shape ignores kinetic energy.

5.4 Mode energy and Parseval's physical meaning

The instantaneous string energy is

$$E = \frac{1}{2}\int_0^L [\mu y_t^2 + \mathcal{T}y_x^2]\mathrm{d}x.$$

For a single fixed–fixed mode $y = q_n(t)\sin k_n x$, sine and cosine integrals both equal $L/2$, giving

$$E_n = \frac{\mu L}{4}(\dot{q}_n^2 + \omega_n^2 q_n^2) = \frac{\mu L}{4}\omega_n^2(A_n^2 + B_n^2).$$

Cross terms vanish between different modes. Consequently $E = \sum_n E_n$ for sufficiently regular finite-energy data. This is Parseval orthogonality expressed as physical energy accounting.

Worked Example: Check a pluck by two independent energies

For the triangular pluck, the slopes are h/a and $-h/(L-a)$. Thus

$$E(0) = \frac{\mathcal{T}}{2}\left(\frac{h^2}{a} + \frac{h^2}{L-a}\right) = \frac{\mathcal{T}h^2 L}{2a(L-a)}.$$

The modal sum must reproduce this value. For $a = L/2$, $E = 2\mathcal{T}h^2/L$. Missing the $1/2$ in the potential-energy density doubles the answer.

5.5 Driven strings and resonance selection

For a transverse force density $f(x)\cos\omega t$ and viscous damping $2\beta\mu y_t$,

$$\mu y_{tt} + 2\beta\mu y_t - \mathcal{T} y_{xx} = f(x)\cos\omega t.$$

The modal force coefficient is $f_n = (2/L)\int_0^L f(x)\sin(k_n x)\mathrm{d}x$, and the steady modal amplitude is

$$\widetilde{q}_n = \frac{f_n/\mu}{\omega_n^2 - \omega^2 - 2\mathrm{i}\beta\omega}.$$

A point drive $F\delta(x - x_d)$ has $f_n = 2F\sin(k_n x_d)/L$. It cannot excite a mode whose node coincides with x_d.

Error Check: What Fourier convergence does and does not say

At an interior jump, the Fourier series converges to the average of the one-sided limits under the usual piecewise-smooth conditions. High-mode truncations can show Gibbs overshoot near a jump. Incompatible nonzero endpoint displacement cannot satisfy a permanently fixed boundary at that endpoint; state the compatibility requirement instead of silently forcing the data to fit.

Standard Workflow: Standing-wave initial-value problems

Derive v, impose endpoint conditions, identify the orthogonal mode basis, project both initial shape and initial velocity, evolve each coefficient, and reconstruct. Verify the initial data and use energy or a symmetry selection rule as an independent check.

6 Traveling Waves, Energy Flux, and Scattering

Source map: Syllabus Unit 5; group-velocity connection to Peatross and Ware Section 7.2.

6.1 d'Alembert's solution and propagation direction

For $y_{tt} = v^2 y_{xx}$ on an infinite string, use $\xi = x - vt$, $\eta = x + vt$. The equation becomes $y_{\xi\eta} = 0$, hence

$$y(x,t) = F(x - vt) + G(x + vt).$$

A point of fixed argument $x - vt$ moves right; a fixed $x + vt$ moves left. A snapshot is not enough to determine propagation direction unless y_t is also supplied.

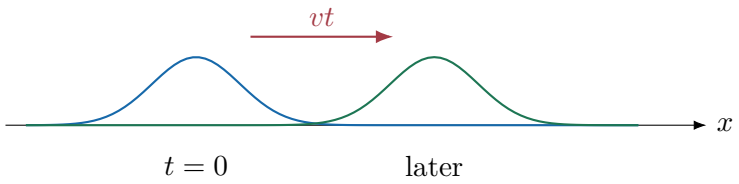


Figure 6.1: A right-moving pulse keeps its shape in a nondispersive medium.

Key Result: The infinite-string initial-value solution

If $y(x,0) = f(x)$ and $y_t(x,0) = g(x)$,

$$y(x,t) = \frac{f(x - vt) + f(x + vt)}{2} + \frac{1}{2v}\int_{x-vt}^{x+vt} g(s)\mathrm{d}s.$$

Derivation: $F + G = f$ and $-vF' + vG' = g$ give $F' = (f' - g/v)/2$, $G' = (f' + g/v)/2$. Integrating and substituting yields the formula. Differentiation at $t = 0$ checks both data.

Worked Example: A released pulse splits, but a launched pulse need not

If $g = 0$, a localized shape f separates into two pulses of half its height. To make a purely right-moving pulse $f(x - vt)$, choose $g(x) = -vf'(x)$. With $f = Ae^{-x^2/(2\sigma^2)}$, the required velocity is $g = vAxe^{-x^2/(2\sigma^2)}/\sigma^2$. Zero initial velocity and purely rightward propagation are incompatible for a nonconstant pulse.

6.2 Energy density and directed power

The transverse kinetic-energy density is $u_K = \mu y_t^2/2$. For small slope the extra length of a segment is $\mathrm{d}\ell - \mathrm{d}x \simeq y_x^2\mathrm{d}x/2$, giving $u_U = \mathcal{T}y_x^2/2$. Differentiate their sum and use the wave equation:

$$\begin{aligned}\partial_t u &= \mu y_t y_{tt} + \mathcal{T}y_x y_{xt} \\ &= \mathcal{T}(y_t y_{xx} + y_x y_{xt}) = \partial_x(\mathcal{T}y_t y_x).\end{aligned}$$

Key Result: Local conservation and string power

$$\partial_t u + \partial_x P = 0, \qquad P = -\mathcal{T}y_x y_t.$$

For $y = A\cos(kx - \omega t)$,

$$\langle P\rangle = \tfrac{1}{2}\mathcal{T}k\omega A^2 = \tfrac{1}{2}Zv_y^2 = \tfrac{1}{2}Z\omega^2 A^2, \qquad Z = \mathcal{T}/v = \sqrt{\mathcal{T}\mu}.$$

Here $v_y = \omega A$ denotes the peak transverse particle speed, not the propagation speed. The mechanical wave impedance Z has units $\mathrm{kg\,s^{-1}}$.

For a right-moving shape, $y_t = -vy_x$ and $P = vu \geq 0$; for a left-moving shape $P = -vu$. For a sum of opposite harmonic waves with complex displacement amplitudes A and B, cross terms cancel from the time-averaged directed flux:

$$\langle P\rangle = \tfrac{1}{2}Z\omega^2(|A|^2 - |B|^2).$$

A standing wave can store energy while carrying zero net average flux.

Worked Example: Recover amplitude from power

For $\mathcal{T} = 40\,\mathrm{N}$, $\mu = 0.010\,\mathrm{kg\,m^{-1}}$, and $f = 20\,\mathrm{Hz}$, $v = 63.25\,\mathrm{m\,s^{-1}}$ and $Z = 0.6325\,\mathrm{kg\,s^{-1}}$. If $\langle P\rangle = 0.50\,\mathrm{W}$, then $A = \sqrt{2\langle P\rangle/(Z\omega^2)} = 0.0100\,\mathrm{m}$. The small-slope check is $kA \simeq 0.0199 \ll 1$; a small displacement relative to an arbitrary meter scale would not be the relevant condition.

6.3 An interface: derive amplitude coefficients from continuity

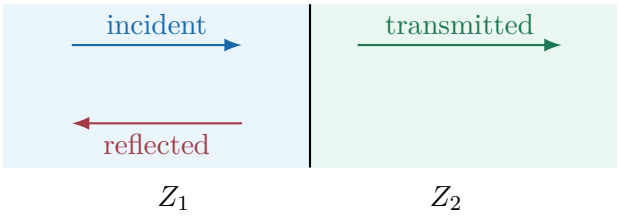


Figure 6.2: The same-frequency incident, reflected, and transmitted waves are matched at a stationary junction.

At $x = 0$, write $\tilde{y}_1 = A_i e^{\mathrm{i}k_1 x} + A_r e^{-\mathrm{i}k_1 x}$ and $\tilde{y}_2 = A_t e^{\mathrm{i}k_2 x}$. For a massless junction, displacement continuity and transverse-force balance give

$$A_i + A_r = A_t, \qquad \mathcal{T}_1 k_1(A_i - A_r) = \mathcal{T}_2 k_2 A_t.$$

Because $\mathcal{T}_j k_j = \omega Z_j$, solution of these two linear equations yields

Key Result: Displacement and power coefficients

$$r = \frac{A_r}{A_i} = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \quad t = \frac{A_t}{A_i} = \frac{2Z_1}{Z_1 + Z_2},$$
$$R = |r|^2, \quad T = \frac{Z_2}{Z_1}|t|^2 = \frac{4Z_1 Z_2}{(Z_1 + Z_2)^2}, \quad R + T = 1.$$

This assumes real positive impedances and a lossless interface. For an ordinary collinear string junction, static tension must also be balanced; differing effective tensions require an appropriate support or waveguide model.

A heavier second string at the same tension has $Z_2 > Z_1$, so $r < 0$: reflection inverts the displacement. In the fixed-end limit $Z_2 \to \infty$, $r \to -1$. In the ideal free-end limit $Z_2 \to 0$, $r \to +1$ and $t \to 2$, yet transmitted power tends to zero. An amplitude greater than the incident amplitude does not violate energy conservation.

Worked Example: An amplitude coefficient is not an energy fraction

Let $\mu_2 = 4\mu_1$ at common tension. Then $Z_2 = 2Z_1$, $r = -1/3$, $t = 2/3$, $R = 1/9$, and $T = 8/9$. The tempting answer $T = 4/9$ fails because the second wave carries more power for a given displacement amplitude.

6.4 A localized barrier: a point mass

A mass M_b joins identical strings of impedance Z. Continuity gives $t = 1 + r$. The net vertical force on the mass is $\mathcal{T}[y_x(0^+) - y_x(0^-)]$. With $\mathrm{e}^{-\mathrm{i}\omega t}$,

$$-M_b\omega^2 A_t = 2\mathrm{i}\omega Z A_r.$$

Therefore

$$r = -\frac{M_b\omega}{M_b\omega + 2\mathrm{i}Z}, \quad t = \frac{2\mathrm{i}Z}{M_b\omega + 2\mathrm{i}Z}, \quad R = \frac{M_b^2\omega^2}{M_b^2\omega^2 + 4Z^2}, \quad T = 1 - R.$$

Low frequencies pass almost freely; high frequencies see an almost fixed boundary. A finite mass stores energy but does not dissipate it. Adding friction would change the energy balance.

Supplementary Background: A finite intermediate layer

Suppose equal exterior impedances Z_1 enclose a length d of impedance Z_2 and wavenumber k_2. Define $r = (Z_1 - Z_2)/(Z_1 + Z_2)$ and $\delta = k_2 d$. The initial transmitted amplitude is $(1 - r^2)\mathrm{e}^{\mathrm{i}\delta}$ times the incident amplitude; each additional round trip multiplies it by $r^2\mathrm{e}^{2\mathrm{i}\delta}$. Summing the geometric series gives

$$t_{\text{layer}} = \frac{(1 - r^2)\mathrm{e}^{\mathrm{i}\delta}}{1 - r^2\mathrm{e}^{2\mathrm{i}\delta}}, \qquad T_{\text{layer}} = \frac{1}{1 + \frac{1}{4}(Z_2/Z_1 - Z_1/Z_2)^2 \sin^2\delta}.$$

The equal exterior impedances make $T = |t|^2$. Perfect transmission occurs at $k_2 d = n\pi$ even when either interface alone reflects. This is resonant cancellation of the total reflected wave, not an absence of internal reflections. It extends the syllabus barrier idea; Problem T8 gives the matching exercise.

6.5 Packets, phase speed, and group speed

For two nearly equal waves, trigonometric addition gives

$$\cos(k_1 x - \omega_1 t) + \cos(k_2 x - \omega_2 t) = 2\cos\frac{\Delta k x - \Delta\omega t}{2}\cos(\bar{k}x - \bar{\omega}t).$$

The carrier moves at $v_p \simeq \bar{\omega}/\bar{k}$ and the envelope at $v_g \simeq \Delta\omega/\Delta k$. For a narrow packet, expand $\omega(k) = \omega_c + v_g(k - k_c) + \frac{1}{2}\omega_c''(k - k_c)^2 + \cdots$. Retaining only the linear term translates the envelope without distortion at $v_g = \omega_c'$. Quadratic and higher terms reshape it.

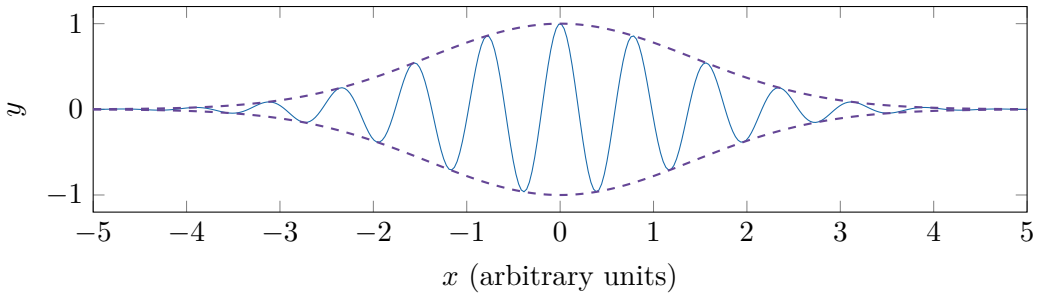


Figure 6.3: A carrier under a smooth envelope. The envelope speed comes from the slope of the dispersion relation.

Worked Example: A dispersive dispersion law

If $\omega = \alpha k^2$ with $k > 0$, then $v_p = \alpha k$ and $v_g = 2\alpha k$. A packet travels twice as fast as its crests to first order. Since $\omega'' = 2\alpha \neq 0$, a finite-bandwidth packet also spreads. A plot of $\omega(k)$ gives v_g from the tangent slope and v_p from the line to the origin.

Standard Workflow: Scattering questions

Write a direction-correct wave on each side; retain the same frequency at a stationary boundary; impose continuity and force balance including any junction mass; solve for complex amplitudes; convert to power using impedance ratios; check $R + T$ and the fixed/free/matched limits.

7 Electromagnetic Waves and Energy Transport

Source map: Syllabus Unit 6; Peatross and Ware Sections 1.6–1.7, 2.1–2.3, 2.6–2.7.

7.1 Maxwell's equations and the material assumptions

For macroscopic fields,

$$\nabla \cdot \boldsymbol{D} = \rho_f, \qquad \nabla \cdot \boldsymbol{B} = 0,$$
$$\nabla \times \boldsymbol{E} = -\partial_t \boldsymbol{B}, \qquad \nabla \times \boldsymbol{H} = \boldsymbol{J}_f + \partial_t \boldsymbol{D}.$$

In a homogeneous isotropic nondispersive linear material, $\boldsymbol{D} = \epsilon \boldsymbol{E}$ and $\boldsymbol{B} = \mu \boldsymbol{H}$. In optics we usually use nonmagnetic media with $\mu = \mu_0$. The polarization is $\boldsymbol{P} = \boldsymbol{D} - \epsilon_0 \boldsymbol{E}$, its bound charge is $\rho_b = -\nabla \cdot \boldsymbol{P}$, and its current is $\boldsymbol{J}_b = \partial_t \boldsymbol{P}$. Material polarization and the polarization state of a wave are different meanings of the same word.

7.2 Derive the wave equation, including the source terms

In vacuum, take the curl of Faraday's law and use Ampere's law:

$$\nabla \times (\nabla \times \boldsymbol{E}) = -\mu_0 \epsilon_0 \partial_t^2 \boldsymbol{E} - \mu_0 \partial_t \boldsymbol{J}.$$

The curl-curl identity and Gauss's law give

$$\nabla^2 \boldsymbol{E} - \frac{1}{c^2} \partial_t^2 \boldsymbol{E} = \mu_0 \partial_t \boldsymbol{J} + \frac{1}{\epsilon_0} \nabla \rho, \qquad c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}.$$

If $\rho_f = 0$ and $\boldsymbol{J} = \boldsymbol{J}_f + \partial_t \boldsymbol{P}$, then

$$\nabla^2 \boldsymbol{E} - \frac{1}{c^2} \boldsymbol{E}_{tt} = \mu_0 \partial_t \boldsymbol{J}_f + \mu_0 \boldsymbol{P}_{tt} - \frac{1}{\epsilon_0} \nabla (\nabla \cdot \boldsymbol{P}).$$

In a source-free homogeneous isotropic material, $\nabla \cdot \boldsymbol{D} = 0$ implies $\nabla \cdot \boldsymbol{E} = 0$, and the nondispersive equation reduces to $\nabla^2 \boldsymbol{E} = \mu\epsilon \boldsymbol{E}_{tt}$. In an inhomogeneous or anisotropic material, $\nabla \cdot \boldsymbol{E} = 0$ cannot simply be assumed.

Key Result: Plane-wave constraints are stronger than the wave equation

For $\boldsymbol{E} = \mathrm{Re}[\boldsymbol{E}_0 \mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}]$ in a source-free isotropic material,

$$k^2 = \mu\epsilon\omega^2, \quad \boldsymbol{k} \cdot \boldsymbol{E}_0 = 0, \quad \boldsymbol{B}_0 = \frac{\boldsymbol{k} \times \boldsymbol{E}_0}{\omega}, \quad \boldsymbol{H}_0 = \frac{1}{Z}\hat{\boldsymbol{k}} \times \boldsymbol{E}_0,$$

$$v_p = \frac{1}{\sqrt{\mu\epsilon}}, \qquad Z = \sqrt{\frac{\mu}{\epsilon}}.$$

The last formula assumes a real propagation direction in a lossless material. In vacuum, $E_0 = cB_0$, $Z_0 \simeq 376.73\,\Omega$. The wave equation alone permits unphysical longitudinal field choices unless Maxwell's divergence constraints are imposed too.

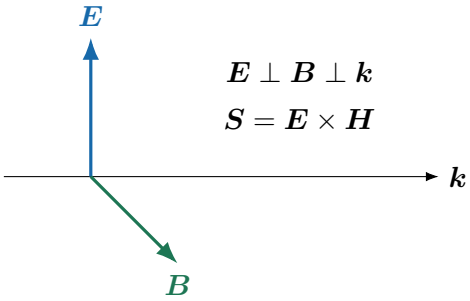


Figure 7.1: The right-handed field relation fixes the direction of energy transport.

Worked Example: Find the accompanying magnetic field correctly

In vacuum, $\boldsymbol{E} = E_0 \cos(kz - \omega t)\hat{\boldsymbol{x}}$. Then $\boldsymbol{B} = (E_0/c)\cos(kz - \omega t)\hat{\boldsymbol{y}}$, since $\hat{\boldsymbol{z}} \times \hat{\boldsymbol{x}} = \hat{\boldsymbol{y}}$. Reversing propagation while keeping the electric polarization makes $\boldsymbol{B}$ reverse sign. A magnetic field in $-\hat{\boldsymbol{y}}$ with the original forward phase would send $\boldsymbol{S}$ backward and violate Faraday's law.

7.3 Poynting's theorem from Maxwell's equations

Dot Ampere's law with $\boldsymbol{E}$ and Faraday's law with $\boldsymbol{H}$, then use the divergence identity:

$$\nabla \cdot (\boldsymbol{E} \times \boldsymbol{H}) = -\boldsymbol{E} \cdot \boldsymbol{J}_f - \boldsymbol{E} \cdot \partial_t \boldsymbol{D} - \boldsymbol{H} \cdot \partial_t \boldsymbol{B}.$$

For real, time-independent, nondispersive ϵ and μ, the last two terms are time derivatives, so

Key Result: Local and integral energy conservation

$$\partial_t u + \nabla \cdot \boldsymbol{S} = -\boldsymbol{J}_f \cdot \boldsymbol{E}, \quad u = \tfrac{1}{2}\epsilon E^2 + \tfrac{1}{2\mu}B^2, \quad \boldsymbol{S} = \boldsymbol{E} \times \boldsymbol{H}.$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V u \,\mathrm{d}V + \oint_{\partial V} \boldsymbol{S} \cdot \hat{\boldsymbol{n}} \,\mathrm{d}A = -\int_V \boldsymbol{J}_f \cdot \boldsymbol{E} \,\mathrm{d}V.$$

Positive outward flux removes field energy; positive $\boldsymbol{J}_f \cdot \boldsymbol{E}$ transfers energy to charges. With polarization treated explicitly, vacuum field energy plus the medium's mechanical energy gives the appropriate balance.

For harmonic fields the exact cycle-average identity is

$$\langle \boldsymbol{S} \rangle = \tfrac{1}{2}\,\mathrm{Re}(\widetilde{\boldsymbol{E}} \times \widetilde{\boldsymbol{H}}^{\,*}).$$

For a single lossless plane wave,

$$I = \frac{|\boldsymbol{E}_0|^2}{2Z} = \tfrac{1}{2}\epsilon v_p |\boldsymbol{E}_0|^2, \quad \langle u \rangle = \tfrac{1}{2}\epsilon |\boldsymbol{E}_0|^2, \quad I = v_p \langle u \rangle .$$

In nonmagnetic material, $Z = Z_0/n$ and $I = \frac{1}{2} n \epsilon_0 c |\boldsymbol{E}_0|^2$. A complex vector norm is $\boldsymbol{E}_0 \cdot \boldsymbol{E}_0^*$; $\boldsymbol{E}_0 \cdot \boldsymbol{E}_0$ is not an intensity.

Worked Example: A beam's electric and magnetic amplitudes

A vacuum beam has power $P = 2.0\,\mathrm{mW}$ uniformly spread over $A = 1.0\,\mathrm{mm}^2$. Thus $I = 2000\,\mathrm{W\,m^{-2}}$, $E_0 = \sqrt{2Z_0 I} = 1.228 \times 10^3\,\mathrm{V\,m^{-1}}$, $B_0 = E_0/c = 4.095\,\mu\mathrm{T}$, and $\langle u \rangle = I/c = 6.671 \times 10^{-6}\,\mathrm{J\,m^{-3}}$. For a Gaussian beam, the stated area would need a specified averaging convention; uniform illumination is part of this model.

7.4 Absorption and complex wave impedance

Use a forward wave with $\widetilde{n} = n + \mathrm{i}\kappa$, $\widetilde{k} = (\omega/c)(n + \mathrm{i}\kappa)$:

$$\widetilde{E}(z) = E_0 \mathrm{e}^{\mathrm{i}n\omega z/c} \mathrm{e}^{-\kappa\omega z/c}.$$

In a nonmagnetic material $\widetilde{H} = \widetilde{n}\widetilde{E}/Z_0$ (with the appropriate perpendicular direction), so E and H are generally out of phase. Their mean flux is

$$I(z) = \frac{n}{2Z_0}|E_0|^2 \mathrm{e}^{-2\kappa\omega z/c} = I(0)\mathrm{e}^{-\alpha z}, \qquad \alpha = \frac{2\kappa\omega}{c} = \frac{4\pi\kappa}{\lambda_0}.$$

The amplitude penetration length is $\delta_A = c/(\kappa\omega)$; the intensity 1/e length is $\delta_I = 1/\alpha = \delta_A/2$.

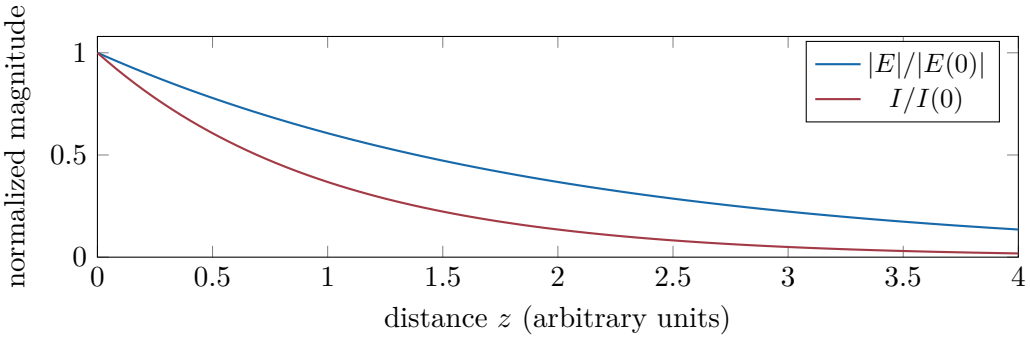


Figure 7.2: An intensity attenuation exponent is twice the field-amplitude exponent.

Worked Example: Absorption measured through two thicknesses

If otherwise identical samples of thicknesses d_1 and d_2 transmit internal intensities I_1 and I_2, and common interface factors cancel, then $\alpha = \ln(I_1/I_2)/(d_2 - d_1)$. For $d_2 - d_1 = 2.0\,\mathrm{mm}$ and $I_2/I_1 = 0.60$, $\alpha = 255.4\,\mathrm{m^{-1}}$. At $\lambda_0 = 600\,\mathrm{nm}$, $\kappa = \alpha\lambda_0/(4\pi) = 1.220 \times 10^{-5}$. Using the external incident intensity without correcting reflection would generally overestimate absorption.

7.5 A useful boundary connection

At normal incidence between lossless dielectrics, tangential E and H continuity give

$$1 + r_E = t_E, \qquad \frac{1 - r_E}{Z_1} = \frac{t_E}{Z_2},$$

$$r_E = \frac{Z_2 - Z_1}{Z_1 + Z_2}, \quad t_E = \frac{2Z_2}{Z_1 + Z_2}, \quad T = \frac{Z_1}{Z_2}|t_E|^2.$$

This is a supplementary bridge to the string interface: the power weighting is inverted because electromagnetic power is $E^2/(2Z)$ whereas string power is $Z\omega^2A^2/2$. It is not safe to copy a string displacement coefficient into an electric-field formula.

Error Check: Energy in a dispersive material

Do not use $u = (\epsilon E^2 + B^2/\mu)/2$ with a complex frequency-dependent ϵ as if it were an instantaneous stored energy. Dispersion introduces material oscillator energy, and loss requires explicit dissipation. Chapter 8 derives the Lorentz-medium balance. For a narrowband wave in a transparent dispersive medium, the cycle-averaged energy is $[\mathrm{d}(\omega\epsilon)/\mathrm{d}\omega]|E_0|^2/4 + \mu_0|H_0|^2/4$ under the stated nonmagnetic, low-loss assumptions.

8 The Lorentz Model: From a Driven Electron to Optical Response

Source map: Syllabus Unit 7; Peatross and Ware Sections 2.3–2.5.

8.1 Microscopic displacement and macroscopic polarization

Let an electron of charge $-e$ and mass m_e have displacement x relative to its positive binding center. The linear model is

$$m_e\ddot{x} + m_e\Gamma\dot{x} + m_e\Omega_0^2 x = -eE(t).$$

The light field is treated as uniform across an atom, the displacement remains small, and magnetic-force effects are negligible for nonrelativistic electron motion. Damping Γ is the coefficient of $\dot{x}$, so it corresponds to 2β in the mechanical chapter. With N identical active oscillators per volume, the dipole density is $P = N(-e)x$.

Key Result: Derive the complex susceptibility

For $E = \mathrm{Re}(E_0\mathrm{e}^{-\mathrm{i}\omega t})$,

$$X = -\frac{eE_0/m_e}{\Omega_0^2 - \omega^2 - \mathrm{i}\Gamma\omega}, \quad P_0 = \epsilon_0\chi E_0,$$

$$\chi(\omega) = \frac{\Omega_p^2}{\Omega_0^2 - \omega^2 - \mathrm{i}\Gamma\omega}, \qquad \Omega_p^2 = \frac{Ne^2}{\epsilon_0 m_e}.$$

The charge appears twice, once in the force and once in the dipole, giving a positive e^2 numerator. The susceptibility is dimensionless.

Let $\Delta = \Omega_0^2 - \omega^2$. Rationalizing gives

$$\chi' = \frac{\Omega_p^2\Delta}{\Delta^2 + \Gamma^2\omega^2}, \qquad \chi'' = \frac{\Omega_p^2\Gamma\omega}{\Delta^2 + \Gamma^2\omega^2}.$$

For a passive oscillator and $\omega > 0$, $\chi'' \geq 0$. The real part controls the in-phase material response; the imaginary part gives the quadrature response that absorbs energy.

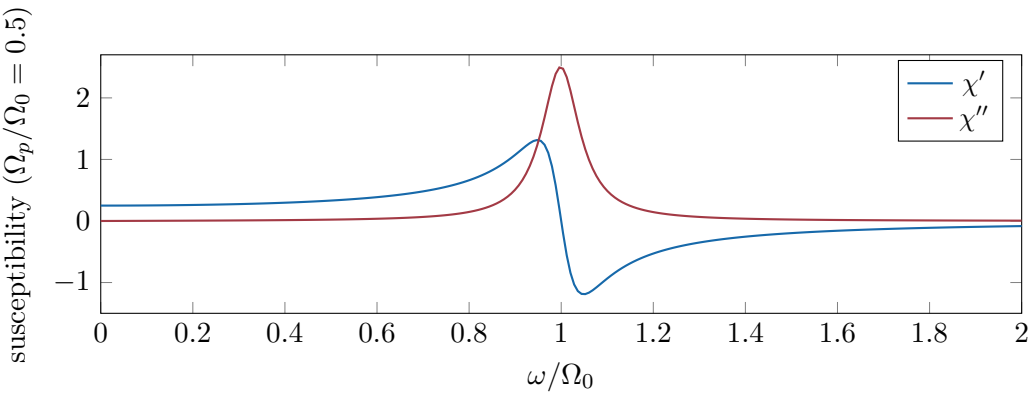


Figure 8.1: Real and imaginary susceptibility for $\Gamma/\Omega_0 = 0.1$. The real part changes sign near the resonance, while the absorptive part is positive.

8.2 From susceptibility to index, phase speed, and absorption

For an isotropic nonmagnetic material, Maxwell's equation gives

$$\widetilde{k}^2 = \frac{\omega^2}{c^2}(1+\chi), \qquad (n+\mathrm{i}\kappa)^2 = 1+\chi'+\mathrm{i}\chi''.$$

Equating real and imaginary parts gives $n^2-\kappa^2 = 1+\chi'$ and $2n\kappa = \chi''$. Let $A = 1+\chi'$, $B = \chi''$, and $R = \sqrt{A^2+B^2}$. Then

Key Result: Exact passive square root and observable quantities

$$n = \sqrt{\frac{R+A}{2}}, \qquad \kappa = \sqrt{\frac{R-A}{2}} \quad (B \geq 0),$$

$$v_p = \frac{c}{n}, \quad \lambda = \frac{\lambda_0}{n}, \quad \alpha = \frac{2\omega\kappa}{c}.$$

When $n > 0$, $\kappa = B/(2n)$ is equivalent. The square-root form also handles $n = 0$ without dividing by zero. If $|\chi| \ll 1$, $n \simeq 1+\chi'/2$ and $\kappa \simeq \chi''/2$. Small absorption alone does not imply small χ'.

Worked Example: Compute an index without a weak-response approximation

Suppose $\chi' = 1.24$ and $\chi'' = 0.30$. Then $A = 2.24$, $R = 2.26$, so $n = 1.50$ and $\kappa = 0.10$. At $\lambda_0 = 600\,\mathrm{nm}$, $\lambda = 400\,\mathrm{nm}$, $\delta_A = \lambda_0/(2\pi\kappa) = 955\,\mathrm{nm}$, and $\delta_I = 477\,\mathrm{nm}$. Checking $2n\kappa = 0.30$ catches the sign and factor-of-two conventions.

8.3 Energy transfer explains the imaginary part

The polarization current amplitude is $J_{P,0} = -\mathrm{i}\omega P_0$. Therefore

$$\langle E J_P \rangle = \frac{1}{2}\,\mathrm{Re}[E_0 J_{P,0}^*] = \frac{1}{2}\epsilon_0\omega\chi''|E_0|^2 \geq 0.$$

Independently, each electron dissipates $m_e\Gamma\langle\dot{x}^2\rangle = m_e\Gamma\omega^2|X|^2/2$. Multiplying by N and inserting X gives the same expression. This verifies both the microscopic model and the phasor sign.

Multiply the microscopic equation by $N\dot{x}$:

$$\partial_t\left[\frac{Nm_e}{2}(\dot{x}^2+\Omega_0^2x^2)\right] = E\dot{P} - Nm_e\Gamma\dot{x}^2.$$

Combining with vacuum-field Poynting conservation yields

$$\partial_t(u_{\mathrm{vac}}+u_{\mathrm{osc}}) + \nabla\cdot\boldsymbol{S} = -Nm_e\Gamma\dot{x}^2.$$

The missing energy becomes heat (or other reservoir energy represented by damping); it has not disappeared. In a monochromatic homogeneous wave, the flux decrease obeys $-\mathrm{d}I/\mathrm{d}z = \alpha I$. Since $\alpha I = (2\omega\kappa/c)(n\epsilon_0 c|E|^2/2) = \epsilon_0\omega\chi''|E|^2/2$, the propagation and oscillator calculations agree exactly.

Worked Example: At the binding resonance

For $\omega = \Omega_0$, $\chi' = 0$ and $\chi'' = \Omega_p^2/(\Gamma\Omega_0)$. This does not force $n = 1$: the exact square root includes χ''^2. Only in the dilute weak-response limit is $n \simeq 1$. Absorbed power has a maximum at Ω_0 for this damping model, but the peak of χ'' itself is slightly shifted because power contains the extra factor ω. Distinguish the measured response before differentiating.

8.4 Low, high, and resonant frequency limits

At $\omega \ll \Omega_0$, $\chi' \simeq \Omega_p^2/\Omega_0^2 > 0$ and $\chi'' \simeq \Omega_p^2\Gamma\omega/\Omega_0^4$. Far above resonance, $\chi' \simeq -\Omega_p^2/\omega^2$ and $\chi'' \simeq \Omega_p^2\Gamma/\omega^3$, so $\widetilde{n} \to 1$. Near resonance, assuming $|\delta|, \Gamma \ll \Omega_0$ with $\delta = \omega - \Omega_0$,

$$\chi \simeq -\frac{\Omega_p^2}{2\Omega_0}\frac{1}{\delta + \mathrm{i}\Gamma/2}.$$

This produces the narrow absorptive line and rapid real-index variation. Normal dispersion means $\mathrm{d}n/\mathrm{d}\omega > 0$; anomalous dispersion means it is negative. Near significant absorption a group velocity computed from a real-index derivative may describe peak reshaping, not a signal or energy velocity.

Supplementary Background: Free-electron and multi-resonance connections

Setting $\Omega_0 = 0$ gives the Drude form $\chi = -\Omega_p^2/(\omega^2 + \mathrm{i}\Gamma\omega)$. In the collisionless limit, $n^2 = 1 - \Omega_p^2/\omega^2$. Above Ω_p waves propagate; below it the bulk solution is evanescent. A decaying evanescent field in a lossless plasma is not dissipative absorption: $\chi'' = 0$ and there is no heating. Multiple bound species contribute $\chi = \sum_j f_j\Omega_{pj}^2/(\Omega_j^2 - \omega^2 - \mathrm{i}\Gamma_j\omega)$. Local-field corrections and microscopic quantum derivations are outside the supplied syllabus.

Worked Example: A transparent plasma separates phase and group speeds

From $\omega^2 = \Omega_p^2 + c^2k^2$, $v_p = \omega/k = c/n$ and $v_g = c^2k/\omega = cn$, hence $v_pv_g = c^2$. At $\omega = 2\Omega_p$, $n = \sqrt{3}/2$, $v_p = 2c/\sqrt{3} > c$, and $v_g = \sqrt{3}c/2 < c$. A superluminal crest speed is not superluminal information transport.

Standard Workflow: Material-response questions

State the phasor convention; solve the microscopic driven equation; multiply by charge and density; separate χ' and χ''; choose the passive square root; calculate wavelength, phase speed, and intensity attenuation; verify $2n\kappa = \chi''$ and positive absorbed power. Use a weak-response approximation only after checking its small parameter.

9 Polarization, Interference, and Wave Packets

Source map: Syllabus Units 6 and 8; Peatross and Ware Sections 6.1–6.6 and 7.1–7.5.

9.1 Polarization is a trajectory, not a propagation direction

For a wave along $+z$, let $\boldsymbol{E}_0 = A\hat{\boldsymbol{x}} + B\mathrm{e}^{\mathrm{i}\delta}\hat{\boldsymbol{y}}$, with $A, B \geq 0$. At a fixed position and after absorbing a common phase, $E_x = A\cos\omega t$ and $E_y = B\cos(\omega t - \delta)$. Eliminating time yields

$$\left(\frac{E_x}{A}\right)^2 + \left(\frac{E_y}{B}\right)^2 - 2\frac{E_x E_y}{AB}\cos\delta = \sin^2\delta.$$

The equation is understood by limits when A or B is zero. If $\delta = 0$ or π, the trajectory is a straight line. If $A = B$ and $\delta = \pm\pi/2$, it is a circle. Otherwise it is generally an ellipse.

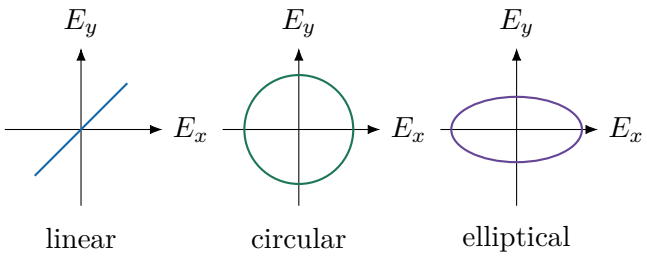


Figure 9.1: The electric-field tip at one position traces a line, circle, or ellipse over one period.

Definition: Jones vectors and the convention for rotation

A normalized Jones vector is $\boldsymbol{e} = (A, B\mathrm{e}^{\mathrm{i}\delta})^T/\sqrt{A^2+B^2}$; multiplication by a common unit complex phase does not change its polarization state. With $\mathrm{e}^{\mathrm{i}(kz-\omega t)}$, the vector $(1,\mathrm{i})^T/\sqrt{2}$ rotates from $+x$ toward $+y$ at fixed $z = 0$ as time increases. Specify this trajectory and viewing direction before using right/left labels, since naming conventions differ. Peatross and Ware call this state left circular.

Worked Example: Determine a polarization and its intensity

For $\boldsymbol{E}_0 = E_a(2,\mathrm{i})^T$, $E_x = 2E_a\cos\omega t$, $E_y = E_a\sin\omega t$ at $z = 0$. Hence $E_x^2/(4E_a^2) + E_y^2/E_a^2 = 1$: an ellipse with axial ratio $1/2$. In a lossless nonmagnetic material, $I = (5/2)n\epsilon_0 c E_a^2$. The common scalar field scale is not the length of the major axis alone.

9.2 Projection and optional matrix tools

An ideal polarizer along real unit vector $\boldsymbol{p} = (\cos\theta, \sin\theta)^T$ produces $\boldsymbol{E}_{\text{out}} = \boldsymbol{p}\boldsymbol{p}^T\boldsymbol{E}_{\text{in}}$. For incident linear polarization at angle α,

$$I_{\text{out}} = I_{\text{in}}\cos^2(\alpha - \theta) \qquad \text{(Malus's law)}.$$

For any coherent polarization, use $I_{\text{out}}/I_{\text{in}} = |\boldsymbol{p}^\dagger\boldsymbol{e}|^2$. Unpolarized light transmits half its intensity through an ideal polarizer; a single Jones vector cannot represent that statistical mixture.

Supplementary Background: Jones matrices connect directly to linear algebra

The polarizer matrix is $P(\theta) = \boldsymbol{p}\boldsymbol{p}^T$, an orthogonal projector with $P^2 = P$. A lossless retarder aligned to x, y is $W = \mathrm{diag}(1, \mathrm{e}^{\mathrm{i}\delta_r})$. For axes rotated by θ, use $W(\theta) = R(\theta)WR(-\theta)$, where $R(\theta) = \begin{pmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{pmatrix}$. The retardance is $\delta_r = (\omega/c)(n_{\text{slow}} - n_{\text{fast}})d$. Quarter- and half-wave plates have $\delta_r = \pi/2$ and π modulo 2π. Multiply device matrices in encounter order from right to left. Retarders preserve vector norm; polarizers generally do not.

Worked Example: A quarter-wave plate makes circular polarization

Send $\boldsymbol{e}_{\text{in}} = (1,1)^T/\sqrt{2}$ through a plate with fast x axis and retardance $\pi/2$. Then $\boldsymbol{e}_{\text{out}} = (1,\mathrm{i})^T/\sqrt{2}$. Its norm remains one, so ideal transmission does not change intensity. Input $(1,0)^T$ would stay linear: a quarter-wave plate does not make every input circular.

9.3 Derive the two-beam interference term

For coherent quasi-parallel waves in the same material, let local complex fields be $\boldsymbol{E}_1$ and $\boldsymbol{E}_2 e^{\mathrm{i}\Delta\phi}$. Using $I = C\boldsymbol{E}\cdot\boldsymbol{E}^*$ with $C = n\epsilon_0 c/2$ gives

$$I = I_1 + I_2 + 2C\,\mathrm{Re}(\boldsymbol{E}_1^* \cdot \boldsymbol{E}_2 e^{\mathrm{i}\Delta\phi}).$$

If $\eta = \boldsymbol{e}_1^\dagger \boldsymbol{e}_2$ for normalized polarization vectors,

Key Result: Visibility and polarization overlap

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\,|\eta|\cos(\Delta\phi + \arg\eta),$$

$$I_{\max,\min} = I_1 + I_2 \pm 2\sqrt{I_1 I_2}|\eta|, \qquad \mathcal{V} = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{2\sqrt{I_1 I_2}|\eta|}{I_1 + I_2}.$$

Full visibility requires equal intensities and identical polarization up to a phase. Orthogonal Jones vectors give no cross term even when both beams are perfectly coherent.

Worked Example: Imbalance and polarization reduce contrast independently

If $I_1 = 9I_0$, $I_2 = I_0$, and the linear polarizations differ by $60°$, then $|\eta| = 1/2$. Therefore $I = 10I_0 + 3I_0\cos\Delta\phi$, $I_{\max} = 13I_0$, $I_{\min} = 7I_0$, and $\mathcal{V} = 0.30$. Equalizing the intensities would raise visibility to 0.50, not to one, because the polarization mismatch remains.

9.4 Spatial fringes and optical path

For plane waves, $\Delta\phi = (\boldsymbol{k}_2 - \boldsymbol{k}_1)\cdot\boldsymbol{r} + \phi_0$. Fringes are planes of constant phase difference perpendicular to $\Delta\boldsymbol{k}$. In the symmetric crossing geometry with total crossing angle 2θ and common k, the transverse fringe spacing is

$$d = \frac{2\pi}{2k\sin\theta} = \frac{\lambda}{2\sin\theta}.$$

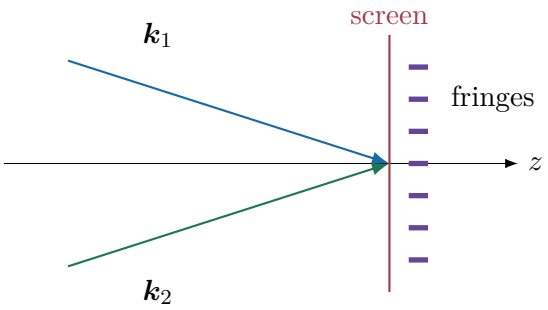


Figure 9.2: Two slightly crossing beams give a transverse sequence of phase-difference maxima.

In a nondispersive path of length ℓ, phase accumulation is $n\omega\ell/c = 2\pi n\ell/\lambda_0$. The optical path difference is $\Delta(\int n\mathrm{d}s)$, with any reflection or device phase added separately.

Worked Example: Count fringe shifts from a path change

In a two-arm arrangement a mirror displacement Δx changes the round-trip path by $2\Delta x$. In air, the phase changes by $4\pi\Delta x/\lambda_0$, and $N = 2\Delta x/\lambda_0$ fringes pass a fixed detector. A $1.50\,\mu$m displacement at $\lambda_0 = 600\,$nm shifts five fringes. The two-pass geometry, not a mysterious extra phase rule, supplies the factor of two.

9.5 Unequal frequencies and the detector timescale

For nearby frequencies in the same polarization,

$$I(x,t) \simeq I_1 + I_2 + 2\sqrt{I_1 I_2}\cos(\Delta k x - \Delta\omega t + \phi_0),$$

after averaging over the rapid optical cycles but not over the beat period. The beat frequency is $|\Delta\omega|/(2\pi)$. A detector averaging over many beat periods sees only $I_1 + I_2$.

For a rectangular exposure of duration T_d centered at t_c, integrating the cosine gives a multiplier

$$\operatorname{sinc}(\Delta\omega T_d/2), \qquad \operatorname{sinc} x = \frac{\sin x}{x}.$$

Its absolute value reduces the measured contrast. Equal-frequency beams with a random relative phase also lose the averaged cross term. Frequency mismatch, polarization orthogonality, and phase randomness are distinct mechanisms.

Worked Example: A beat can exist but disappear from the measurement

Two equal-polarization beams differ by 1 kHz. A 1 ms exposure spans exactly one beat period, so $\Delta\omega T_d/2 = \pi$ and $\operatorname{sinc}\pi = 0$. A sufficiently fast photodiode resolves the oscillation, but that integrating detector records the constant sum. Neither conclusion contradicts coherent field superposition.

9.6 Group index, delay, and pulse spreading

In a transparent medium, $k(\omega) = n(\omega)\omega/c$. Therefore

$$v_g = \left(\frac{\mathrm{d}k}{\mathrm{d}\omega}\right)^{-1} = \frac{c}{n + \omega n'}, \qquad n_g = n + \omega n' = n - \lambda_0\frac{\mathrm{d}n}{\mathrm{d}\lambda_0}.$$

The last equality follows from $\omega = 2\pi c/\lambda_0$. A slab of thickness d gives group delay $\tau_g = n_g d/c$, and extra delay relative to vacuum $(n_g - 1)d/c$. Phase delay nd/c is generally different.

For a narrowband amplitude $A(0,t) = A_0 \mathrm{e}^{-t^2/(2\tau_0^2)}$, define $k_2 = \mathrm{d}^2k/\mathrm{d}\omega^2$ and $\tau = t - k_0' z$. Its Gaussian spectrum, in the detuning $\Omega = \omega - \omega_c$, is proportional to $\mathrm{e}^{-\tau_0^2\Omega^2/2}$. After factoring out the carrier phase, second-order propagation therefore requires

$$A(z,t) = \frac{A_0\tau_0}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\exp\left[-\frac{\tau_0^2 - \mathrm{i}k_2 z}{2}\Omega^2 - \mathrm{i}\tau\Omega\right]\mathrm{d}\Omega.$$

Complete the square, or use $\int_{-\infty}^{\infty}\mathrm{e}^{-a\Omega^2+b\Omega}\mathrm{d}\Omega = \sqrt{\pi/a}\mathrm{e}^{b^2/(4a)}$ for $\operatorname{Re} a > 0$. With $a = (\tau_0^2 - \mathrm{i}k_2 z)/2$ and $b = -\mathrm{i}\tau$, this gives

$$A(z,t) = \frac{A_0}{\sqrt{1 - \mathrm{i}k_2 z/\tau_0^2}}\exp\left[-\frac{\tau^2}{2\tau_0^2(1 - \mathrm{i}k_2 z/\tau_0^2)}\right].$$

The intensity width parameter becomes $\tau(z) = \tau_0\sqrt{1 + (k_2 z/\tau_0^2)^2}$. This is a supplementary quantitative extension of the syllabus group-velocity idea. It assumes negligible frequency-dependent loss and sufficiently small higher-order dispersion over the pulse spectrum.

Worked Example: Group delay from wavelength data

At $\lambda_0 = 500\,\text{nm}$ let $n = 1.50$ and $\mathrm{d}n/\mathrm{d}\lambda_0 = -0.040\,\mu\text{m}^{-1}$. Converting λ_0 to $0.500\,\mu\text{m}$, $n_g = 1.520$. A $2.00\,\text{cm}$ slab gives $\tau_g = 101.4\,\text{ps}$ and an excess delay $34.7\,\text{ps}$. Keeping wavelength and derivative units consistent is essential.

9.7 Electromagnetic standing waves: fields and flux

Superpose equal x-polarized waves moving in opposite z directions in vacuum. Their magnetic fields have opposite signs, so

$$E_x = 2E_a \cos kz \cos \omega t, \qquad B_y = \frac{2E_a}{c} \sin kz \sin \omega t.$$

Electric and magnetic antinodes are displaced by $\lambda/4$. The instantaneous flux is

$$S_z = \frac{E_a^2}{\mu_0 c} \sin(2kz) \sin(2\omega t), \qquad \langle S_z \rangle = 0.$$

Energy flows locally back and forth. The average total energy density is $\epsilon_0 E_a^2$, independent of z, even though the electric part alone is proportional to $\cos^2 kz$. The local electric-field-squared response used in optical interference is therefore not the same as net directed flux.

Error Check: Where an interference formula applies

The scalar formula $I \propto |E_1 + E_2|^2$ is a field-strength detector model for comparable, quasi-parallel waves. For arbitrary crossing angles, magnetic fields and the detector surface normal must be included through $\langle \boldsymbol{S} \rangle \cdot \hat{\boldsymbol{n}}$. For counterpropagating waves, compute net flux separately; it is not the standing-wave electric intensity pattern.

10 Integrated Revision and Formula Map

10.1 One mathematical structure across the course

Model	Inertia	Restoring term	Response problem
single mass	m	k	$k - m\omega^2 - 2\mathrm{i}m\beta\omega$
coupled masses	M	K	$K - \omega^2 M - \mathrm{i}\omega C$
string	μ	$-\mathcal{T}\partial_x^2$	boundaries select k_n
EM field	ϵ	curl structure	$k^2 - \mu\epsilon\omega^2$
bound electron	m_e	$m_e\Omega_0^2$	$\Omega_0^2 - \omega^2 - \mathrm{i}\Gamma\omega$

Eigenvalue problems describe free motion. Inverting the driven operator describes forced motion. Orthogonality projects initial conditions and forcing into independent channels. Boundary conditions and constitutive laws decide which mathematical solutions are physical.

10.2 Derivation checklist by syllabus unit

Unit	Be able to derive, not just quote
1	Equilibrium curvature; initial-value motion in all damping regimes; phasor response; amplitude and power resonance; cycle-averaged power; linewidth and quality factor.

Unit	Be able to derive, not just quote
2	Spring extensions and stiffness matrix; symmetric/antisymmetric modes; mass orthogonality; beats; transverse coupling from tension and small slopes; forcing selection.
3	Monatomic dispersion; allowed k from boundaries; sound limit; binary determinant, both branches, gap and equal-mass limit; modal forcing.
4	String force balance; endpoint eigenfunctions; Fourier coefficients for shape and velocity; triangular pluck; modal energy; point-drive selection.
5	d'Alembert solution; energy continuity; impedance; interface and point-mass scattering; power conservation; v_p versus v_g.
6	Maxwell curl-curl derivation; transverse plane-wave constraints; E, B, H, k relations; Poynting conservation; average irradiance; attenuation and penetration lengths.
7	Electron displacement to polarization and susceptibility; real/imaginary parts; exact n, κ; heat production; dispersion and frequency limits.
8	Polarization ellipse; vector interference term; contrast; spatial fringe phase; finite-frequency beats and averaging; group delay; standing EM field and zero net average flux.

10.3 A concise equation index

Key Result: Dynamics and modes

$$\omega_0^2 = k/m, \quad \omega_d^2 = \omega_0^2 - \beta^2, \quad \omega_{\text{amp}}^2 = \omega_0^2 - 2\beta^2 \quad \text{when positive},$$
$$Q = \omega_0/(2\beta), \quad \overline{P} = m\beta\omega^2 A^2, \quad K\boldsymbol{a} = \omega^2 M\boldsymbol{a},$$
$$\omega_{\text{mono}}^2 = 4K\sin^2(ka/2)/m, \quad v_g = \omega'(k), \quad v_p = \omega/k.$$

Key Result: Strings and interfaces

$$v = \sqrt{\mathcal{T}/\mu}, \quad Z = \sqrt{\mathcal{T}\mu}, \quad \omega_n = n\pi v/L,$$
$$P = -\mathcal{T}y_x y_t, \quad \langle P\rangle = Z\omega^2 A^2/2,$$
$$r_y = (Z_1 - Z_2)/(Z_1 + Z_2), \quad t_y = 2Z_1/(Z_1 + Z_2), \quad T_y = (Z_2/Z_1)|t_y|^2.$$

Key Result: Light, matter, and interference

$$\boldsymbol{B}_0 = \boldsymbol{k}\times\boldsymbol{E}_0/\omega, \quad \langle\boldsymbol{S}\rangle = \tfrac{1}{2}\,\mathrm{Re}(\boldsymbol{E}_0\times\boldsymbol{H}_0^*),$$
$$\chi = \frac{\Omega_p^2}{\Omega_0^2 - \omega^2 - \mathrm{i}\Gamma\omega}, \quad (n+\mathrm{i}\kappa)^2 = 1+\chi,$$
$$\alpha = 2\omega\kappa/c, \quad v_p = c/n, \quad v_g = c/(n+\omega n'),$$
$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\,\mathrm{Re}(\eta e^{\mathrm{i}\Delta\phi}), \quad \mathcal{V} = \frac{2\sqrt{I_1 I_2}|\eta|}{I_1 + I_2}.$$

10.4 An integrated worked application

Worked Example: Infer a material response from phase and transmission

A nonmagnetic uniform slab has internal wavelength 400 nm for vacuum wavelength 600 nm. Its internal intensity falls to e^{-1} after $d = 1.00\,\text{mm}$. Determine the complex index, susceptibility, phase speed, and heating rate per volume at a point where $I = 100\,\text{W m}^{-2}$.

Recognition and approach. Wavelength fixes the real index; transmission fixes α, hence κ; susceptibility follows by squaring, not by identifying χ with $n - 1$.

First $n = \lambda_0/\lambda = 1.50$. Next $\alpha = 1/d = 1000\,\text{m}^{-1}$, so $\kappa = \alpha\lambda_0/(4\pi) = 4.775 \times 10^{-5}$. Therefore

$$\chi' = n^2 - \kappa^2 - 1 \simeq 1.25, \qquad \chi'' = 2n\kappa = 1.432 \times 10^{-4}.$$

The phase speed is $c/1.5 = 1.999 \times 10^8\,\text{m s}^{-1}$ and the local heating rate is $\alpha I = 1.00 \times 10^5\,\text{W m}^{-3}$. Group speed cannot be determined from one measured index without frequency-derivative information. If the quoted transmission were external rather than internal, surface reflections would also need to be separated.

10.5 A complete, assessable answer

Standard Workflow: The six lines every long solution needs

1. Define coordinates, directions, amplitudes, and assumptions.
2. Derive the governing equation or boundary equations from physical laws.
3. Identify the method: initial-value solution, eigenmodes, Fourier projection, phasor inversion, or scattering match.
4. Show the decisive intermediate algebra or integral; separate singular or zero-parameter cases.
5. State the requested answer with units, phases, and subquestion labels.
6. Verify an initial condition, conservation law, sign, or limiting case that could expose an error.

Error Check: What cannot be predicted from these sources

The syllabus specifies required knowledge, not an examination distribution. The supplied optical textbook provides examples and exercises, but no past paper for this specific course. The practice set therefore labels all questions as original course-aligned training. Topic priority here means structural importance or conceptual difficulty, not a claimed statistical frequency in the school's examinations.

10.6 Source and production record

Subject sources inspected: `syllabus_waves.pdf`, two pages; Peatross and Ware, *Physics of Light and Optics*, 2015 edition, May 15, 2024 revision, 346 PDF pages. Only syllabus-relevant sections and explicitly marked short extensions are used. The supplied Algebra 1E V3.0 notes determine the visual style, not the subject scope. All diagrams are newly drawn vector illustrations. Equations use a single negative-time-exponential convention. The matching practice volume covers every syllabus unit with eight multipart problems and complete inline worked solutions.